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M3dblob functions

Synopsis

The functions prefixed with M3dblob make up the Aurora Imaging Library 3D Blob module.

Functions

M3dblobAlloc
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M3dblobAlloc

Synopsis

Allocate a 3D blob analysis context.

Syntax

```
AIL_ID M3dblobAlloc(
    AIL_ID SysId,              //in
    AIL_INT64 ContextType,     //in
    AIL_INT64 ControlFlag,     //in
    AIL_ID *Context3dblobIdPtr //out
)
```

Description

This function allocates a 3D blob analysis context on the specified system. A 3D blob analysis context contains information needed to perform a M3dblobCalculate() or M3dblobSegment() operation. You can also use **M3dblobAlloc**() to allocate a draw 3D blob analysis context for drawing results using M3dblobDraw3d().

When the 3D blob analysis context is no longer required, release it using M3dblobFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the 3D blob analysis context's lifetime and you must not manually free it.

Parameters

SysId

Specifies the identifier of the system on which to allocate the 3D blob analysis context. Set this parameter to one of the values below:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ContextType

Specifies the type of 3D blob analysis context to allocate. Set this parameter to one of the values below:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of 3D blob analysis context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_CALCULATE_CONTEXT	Allocates a calculate 3D blob analysis context for use with M3dblobCalculate().	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_3D_CONTEXT	Allocates a draw 3D blob analysis context for use with M3dblobDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_SEGMENTATION_CONTEXT	Allocates a segmentation 3D blob analysis context for use with M3dblobSegment().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Context3dblobIdPtr

Specifies the address of the variable in which to write the 3D blob analysis context identifier or specifies the data type that the function should use to return the 3D blob analysis context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D blob analysis context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D blob analysis context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DBLOB_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D blob analysis context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b

Address in which to write the calculate 3D blob analysis context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated calculate 3D blob analysis context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the draw 3D blob analysis context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated draw 3D blob analysis context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the segmentation 3D blob analysis context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated segmentation 3D blob analysis context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 3D blob analysis context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DBLOB_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobAllocResult

Synopsis

Allocate a 3D blob analysis result buffer.

Syntax

```
AIL_ID M3dblobAllocResult(
    AIL_ID SysId,             //in
    AIL_INT64 ResultType,     //in
    AIL_INT64 ControlFlag,    //in
    AIL_ID *Result3dblobIdPtr //out
)
```

Description

This function allocates a 3D blob analysis result buffer on the specified system to store results obtained from an M3dblobCalculate() or M3dblobSegment() operation.

When the 3D blob analysis result buffer is no longer required, release it using M3dblobFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the 3D blob analysis result buffer's lifetime and you must not manually free it.

Parameters

SysId

Specifies the system on which to allocate the result buffer. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result buffer to allocate. Set this parameter to the value below:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of result buffer to allocate																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_SEGMENTATION_RESULT	Specifies to allocate a 3D blob analysis result buffer used to store M3dblobSegment() results.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

Result3dblobIdPtr

Specifies the address of the variable in which to write the 3D blob analysis result buffer identifier or specifies the data type that the function should use to return the 3D blob analysis result buffer identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D blob analysis result buffer ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D blob analysis result buffer ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DBLOB_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D blob analysis result buffer (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the segmentation 3D blob analysis result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated segmentation 3D blob analysis result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 3D blob analysis result buffer identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DBLOB_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.

Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobCalculate

Synopsis

Perform 3D blob analysis calculations.

Syntax

```
void M3dblobCalculate(
    AIL_ID CalculateContext3dblobId, //in
    AIL_ID ContainerBufId,           //in
    AIL_ID Result3dblobId,           //out
    AIL_INT64 LabelOrIndex,          //in
    AIL_INT64 ControlFlag            //in
)
```

Description

This function calculates the specified features for the specified blobs. The features are calculated from the blobs in the source point cloud container, and results are added to the specified result buffer, which is the result buffer that holds the blob segmentation computed from the same source point cloud. To specify the features to calculate, pass a 3D blob analysis context that has the required features enabled for calculation. Use M3dblobControl() to enable the features. Use M3dblobSegment() to fill the result buffer with blob identification information; the result buffer is used to identify which points in the specified container belong to the same blob. If you want to calculate features for a subset of the blobs in the point cloud, use M3dblobSelect() and M3dblobCombine() to fill a result buffer with blob identifying information for the required blobs.

M3dblobCalculate() is optimized to not recalculate the same features twice. If you call the function several times without modifying the container's range component, **M3dblobCalculate**() only calculates features that are enabled in the context but not yet stored in the result buffer. If the range component has changed, the function removes all feature results from the result buffer and then calculates the features enabled in the context.

Parameters

CalculateContext3dblobId

Specifies the identifier of the 3D blob analysis context that specifies the feature(s) to calculate, and other settings related to the 3D blob analysis operation. The context must have been previously allocated using M3dblobAlloc() with M_CALCULATE_CONTEXT.

ContainerBufId

Specifies the identifier of the 3D-processable point cloud container from which the blobs originate. This must be the same container previously passed to M3dblobSegment(), and its confidence component must not have been modified since then. If the segmentation was performed using a label image instead, this can be any container as long as the size of its range component is the same as the label image.

The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must be 3D-processable (that is, if you call MbufInquireContainer() with M_3D_PROCESSABLE, the function returns M_PROCESSABLE). Note that the container must not be a child container on a Distributed Aurora Imaging Library remote system with a parent that contains multiple range or confidence components; otherwise, an error is generated.

Result3dblobId

Specifies the identifier of the 3D blob analysis result buffer that holds the blob segmentation, and in which to store results. The result buffer must have been previously allocated using M3dblobAllocResult().

LabelOrIndex

Specifies the label or index of the blob(s) for which to calculate features, or specifies that you are calculating features

for all blobs. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the blob label or index																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_BLOB_INDEX(AIL_INT Index)	Specifies the index of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Specifies the index.															
	0 <= *Value* < M_NUMBER Specifies the valid index of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_LABEL(AIL_INT Label)	Specifies the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Label Specifies the label.															
	1 <= *Value* <= M_MAX_LABEL_VALUE Specifies the valid label of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL_BLOBS	Specifies to calculate features for all blobs.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobCombine

Synopsis

Combine sets of blobs from different result buffers into one result buffer.

Syntax

```
void M3dblobCombine(
    AIL_ID Src1Result3dblobId, //in
    AIL_ID Src2Result3dblobId, //in
    AIL_ID DstResult3dblobId,  //out
    AIL_INT64 Operation,       //in
    AIL_INT64 ControlFlag      //in
)
```

Description

This function combines sets of blobs from different 3D blob analysis result buffers into one result buffer, according to the specified combine operation. The two source result buffers must hold sets of blob identifying information and feature results from two different selection operations (M3dblobSelect()). The subsets in the source result buffers must also originate from the same blob segmentation (M3dblobSegment()).

M3dblobCombine() allows you to perform compound selections. For example, to select blobs that meet criterion 1 OR criterion 2, first call M3dblobSelect() twice with two different 3D blob analysis result buffers: once to select for criterion 1, and then a second time to select for criterion 2. Then, combine the two sets of blob identifying information and feature results using **M3dblobCombine**() with M_OR. Note that different feature results for the same blob in two source result buffers are combined and assigned to that same blob in the destination result buffer. For example, if source 1 has blob 42's feature A result and source 2 has blob 42's feature B result, then the destination result buffer will have both feature A and feature B results for blob 42 (provided the specified combine operation doesn't exclude blob 42). An error is reported if both sources contain the same feature's results, but the feature's values are different.

To merge two blobs into one blob, use M3dblobControl() with M_MERGE.

This function supports in-place processing; you can specify the identifier of the same result buffer for one of the sources and the destination.

Parameters

Src1Result3dblobId

Specifies the identifier of the first source 3D blob analysis result buffer, previously allocated using M3dblobAllocResult() with M_SEGMENTATION_RESULT.

Src2Result3dblobId

Specifies the identifier of the second source 3D blob analysis result buffer, previously allocated using M3dblobAllocResult() with M_SEGMENTATION_RESULT.

Note that the subset of blobs in this result buffer must originate from the same blob segmentation as the subset of blobs in the Src1Result3dblobId result buffer.

DstResult3dblobId

Specifies the identifier of the 3D blob analysis result buffer in which to store the results of the combine operation. In-place processing is supported; DstResult3dblobId can refer to the same result buffer as Src1Result3dblobId or Src2Result3dblobId.

The result buffer must have been previously allocated using M3dblobAllocResult() with M_SEGMENTATION_RESULT.

Operation

Specifies how to combine the sets of blobs. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying how to combine the sets of blobs																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_AND	Selects blobs for which selected feature results exist in both source result buffers. This is an intersection, equivalent to *Src1* ∩ *Src2* in set language notation.	c		e	f	g	a		i		j	k	l	m	n	b
M_OR	Selects blobs for which selected feature results exist in any source result buffer. This is a union, equivalent to *Src1* ∪ *Src2* in set language notation.	c		e	f	g	a		i		j	k	l	m	n	b
M_SUB	Selects blobs for which selected feature results exist in the first source result buffer, but not the second. This is a subtraction, equivalent to *Src1* - *Src2* in set language notation.	c		e	f	g	a		i		j	k	l	m	n	b
M_XOR	Selects blobs for which selected feature results exist in one source result buffer, but not both. This is a symmetric difference, equivalent to (*Src1* ∪ *Src2*) - (*Src1* ∩ *Src2*) in set language notation.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobControl

Synopsis

Control a 3D blob analysis context or result buffer setting.

Syntax

```
void M3dblobControl(
    AIL_ID ContextOrResult3dblobId, //out
    AIL_INT64 LabelOrIndex,         //in
    AIL_INT64 ControlType,          //in
    AIL_DOUBLE ControlValue         //in
)
```

Description

This function controls the settings of a 3D blob analysis context or result buffer.

You can inquire most of these settings using M3dblobInquire().

To control draw 3D blob analysis settings for M3dblobDraw3d(), use M3dblobControlDraw() instead.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **M3dblobControl()** internally calls the *AIL_DOUBLE* version of this function (M3dblobControlDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (M3dblobControlInt64()).

Parameters

ContextOrResult3dblobId

Specifies the identifier of the 3D blob analysis context or result buffer to control. The 3D blob analysis context or result buffer must have been previously allocated on the system using M3dblobAlloc() or M3dblobAllocResult(), respectively.

See the Parameter associations section for possible values that can be returned.

LabelOrIndex

Specifies the label or index of the blob to control, or specifies to control all blobs. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the blob label or index																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default value. If a 3D blob analysis context is specified, same as M_CONTEXT. If a 3D blob analysis result buffer is specified, same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_INDEX(AIL_INT Index)	Specifies the index of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

	Index Specifies the index; the first blob is at index 0.															
	0 <= *Value* < M_NUMBER Specifies the valid index of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_LABEL(AIL_INT Label)	Specifies the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Label Specifies the label; labeling begins at 1.															
	1 <= *Value* <= M_MAX_LABEL_VALUE Specifies the valid label of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL_BLOBS	Specifies to apply the control setting to all blobs.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT	Specifies to control a general setting of a 3D blob analysis context.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to control a general setting of a 3D blob analysis result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

ControlType

Specifies the setting to change.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the setting's new value.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The table below lists possible values for the **ControlType** and **ControlValue** parameters and possible values returned to the **ContextOrResult3dblobId** parameter.

- For specifying the control type and control value for a 3D blob analysis context or result buffer

The following ContextOrResult3dblobId, ControlType, and ControlValue parameter settings can be specified for different types of 3D blob analysis contexts or result buffers.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the control type and control value for a 3D blob analysis context or result buffer

ContextOrResult3dblobId - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlType																
ControlValue																
Calculate 3D blob analysis context ID	Specifies a calculate 3D blob analysis context. allocated using M3dblobAlloc() with M_CALCULATE_CONTEXT, and used in M3dblobCalculate() operations. In this case, you must set LabelOrIndex to M_CONTEXT or M_DEFAULT.															
M_ALL_FEATURES	Sets whether to calculate all blob features. You can then selectively enable or disable specific features for calculation. It is not recommended to use M_ALL_FEATURES in your final application. This is because programs using M_ALL_FEATURES set to M_ENABLE might become slower when new Aurora Imaging Library versions are released, if new features have been added.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies to disable the calculation of all features.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to enable the calculation of all features.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOUNDING_BOX	Sets whether to calculate the axis-aligned bounding box and its related Ferets. You can retrieve results for these features using M3dblobGetResult() with M_BOX_CENTER_..., M_FERET_X, M_FERET_Y, M_FERET_Z, M_MAX_..., M_MIN_..., and M_SIZE_.... INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the axis-aligned bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the axis-aligned bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID	Sets whether to calculate the centroid. This is equivalent to enabling M_MOMENTS with an order of at least 1. You can retrieve results for this feature using M3dblobGetResult() with M_CENTROID_.... INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the centroid.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the centroid.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_CONTACT_POINTS	Sets whether to save the contact points at each Feret; the contact points are the coordinates of the Feret diameter's start and end points. You can retrieve results for these features using M3dblobGetResult() with M_FERET_... and M_FERET_CONTACT_POINTS_.... INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to save the contact points at each Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to save the contact points at each Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL +	Sets whether to calculate the Feret along a custom direction specified with M_FERET_GENERAL + M_FERET_DIRECTION_.... You can retrieve results for this feature using M3dblobGetResult() with M_FERET_GENERAL. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the Feret along a custom direction.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the Feret along a custom direction. > Note that you must specify the direction vector with M_FERET_DIRECTION_....	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER	Sets whether to calculate the longest Feret. Note that the Feret's direction is accurate up to the angle specified with M_FERET_PRECISION. You can retrieve results for this feature using M3dblobGetResult() with M_FERET_MAX_DIAMETER. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the longest Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the longest Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_DIAMETER	Sets whether to calculate the shortest Feret. Note that the Feret's direction is accurate up to the angle specified with M_FERET_PRECISION. You can retrieve results for this feature using M3dblobGetResult() with M_FERET_MIN_DIAMETER. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to calculate the shortest Feret.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to calculate the shortest Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_PRECISION		Sets the accuracy when calculating the minimum and maximum Feret directions. The smaller the specified angle, the longer the calculation will take. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 20.0 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* < 90.0	Specifies the precision (in degrees).	c		e	f	g	a		i		j	k	l	m	n	b
M_LINEARITY		Sets whether to calculate the degree to which the blobs are linear. This is equivalent to enabling M_MOMENTS with an order of at least 2. You can retrieve results for this feature using M3dblobGetResult() with M_LINEARITY. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to calculate the linearity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to calculate the linearity.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_ORDER		Sets the order up to which moments are calculated. For example, setting this to 3 will calculate third order moments, but also second and first order moments. Note that there are (N+1)(N+2)/2 moments for order N. For example, if M_MOMENT_ORDER is set to 3, the number of computed moments are 10, 6, and 3 for third, second, and first order moments, respectively. See M_MOMENTS for the moments feature results that are available, depending on the order specified with M_MOMENT_ORDER. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 2.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the moment order.	c		e	f	g	a		i		j	k	l	m	n	b

M_MOMENTS	Sets whether to calculate all moments up to M_MOMENT_ORDER. You can retrieve results for these features using M3dblobGetResult() with M_MOMENT_CENTRAL_XYZ(), and M_MOMENT_XYZ(). If the moment order is set to at least 1, you can also retrieve M_CENTROID_... results; if the moment order is set to at least 2, you can also retrieve M_COVARIANCE_MATRIX, M_EIGENVALUE_..., M_LINEARITY, M_PLANARITY, and M_PRINCIPAL_AXIS_... results. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the moments.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the moments.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB +	Sets whether to calculate features related to the nearest blob. You can retrieve results for these features using M3dblobGetResult() with M_NEAREST_BLOB, M_NEAREST_BLOB_DISTANCE, and M_NEAREST_POINT_.... INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate features related to the nearest blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate features related to the nearest blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_PCA	Sets whether to calculate the principal components, which are used in a principal component analysis (PCA) or when calculating M_PCA_BOX. This is equivalent to enabling M_MOMENTS with an order of at least 2. You can retrieve results for these features using M3dblobGetResult() with M_PRINCIPAL_AXIS_.... INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the principal components.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the principal components.	c		e	f	g	a		i		j	k	l	m	n	b

M_PCA_BOX	Sets whether to calculate the PCA-aligned bounding box and its related Ferets. You can retrieve results for these features using M3dblobGetResult() with M_BOX_CENTER_..., M_FERET_AT_PRINCIPAL_AXIS_..., and M_SIZE_.... The PCA-aligned bounding box is typically a tighter fit than other boxes (for example, M_BOUNDING_BOX) for highly planar or linear objects. Note that enabling M_PCA_BOX automatically enables M_PCA. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the PCA-aligned bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the PCA-aligned bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANARITY	Sets whether to calculate the degree to which the blobs are planar. This is equivalent to enabling M_MOMENTS with an order of at least 2. You can retrieve results for this feature using M3dblobGetResult() with M_PLANARITY. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the planarity.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the planarity.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX	Sets whether to calculate the semi-oriented bounding box and its related Ferets. You can retrieve results for these features using M3dblobGetResult() with M_BOX_CENTER_..., M_FERET_SEMI_ORIENTED_HEIGHT, M_FERET_SEMI_ORIENTED_WIDTH, M_FERET_Z, M_MAX_Z, and M_MIN_Z. The semi-oriented bounding box is the minimum volume box that is axis-aligned in Z but not in X and Y. This box is useful when the objects lay flat in 1 dimension (for example, on a conveyor). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the semi-oriented bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the semi-oriented bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b

Segmentation 3D blob analysis context ID	Specifies a segmentation 3D blob analysis context. allocated using M3dblobAlloc() with M_SEGMENTATION_CONTEXT, and used in M3dblobSegment() operations. In this case, you must set LabelOrIndex to M_CONTEXT or M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DISTANCE_MAX	Sets the maximum difference between the color of two points for them to be considered neighbors. Set how to calculate the distance with M_COLOR_DISTANCE_MODE. This control type is available when the component M_COMPONENT_REFLECTANCE or M_COMPONENT_INTENSITY exists in the source container. If you set a non-infinite value, and neither a reflectance nor intensity component exists, am Aurora Imaging Library error is reported. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies not to consider the color.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum color difference.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DISTANCE_MODE	Sets how to calculate the difference in color between two points. If the intensity or reflectance component is monochrome, both modes calculate the same color difference. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_EUCLIDEAN.	c		e	f	g	a		i		j	k	l	m	n	b
M_EUCLIDEAN	Specifies to use the Euclidean distance to calculate the color difference. A Euclidean distance is the square root of the sum of the squared differences between the color of each point's corresponding element in its intensity or reflectance component's 2D grid organization.	c		e	f	g	a		i		j	k	l	m	n	b
M_MANHATTAN	Specifies to use the Manhattan distance to calculate the color difference. A Manhattan distance is the sum of the absolute value of the differences between the color of each point's corresponding element in its intensity or reflectance component's 2D grid organization.	c		e	f	g	a		i		j	k	l	m	n	b
M_CUSTOM_DISTANCE_COMPONENT	Sets to use an extra component during the segmentation. The custom component must be 1-band and have the same size as the range component. Points are considered neighbors if their respective values in the custom component differ by less than M_CUSTOM_DISTANCE_MAX. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_NULL	Specifies that there is no custom component. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_BY_ID(AIL_INT ID)	Specifies to use the component, in the container, that has the specified buffer identifier. An error will be generated if the container does not have a component with this buffer identifier.	c		e	f	g	a		i		j	k	l	m	n	b
			Parameters															
			ID Specifies the Aurora Imaging Library identifier of the component to control.															
		M_COMPONENT_BY_INDEX(AIL_INT ComponentIndex)	Specifies to use the component, in the container, at the specified index. An error will be generated if the container does not have a component at this index.	c		e	f	g	a		i		j	k	l	m	n	b
			Parameters															
			ComponentIndex Specifies the index of the component to control.															
		M_COMPONENT_CONFIDENCE	Specifies that the component stores confidence information for the M_COMPONENT_RANGE or M_COMPONENT_DISPARITY component of the container. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_COORDINATE_MAP_A_AIL	Specifies that the component stores the A coordinate map provided by the camera. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_COORDINATE_MAP_B_AIL	Specifies that the component stores the B coordinate map provided by the camera. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_CUSTOM + *n*	Specifies that the component has a custom component type, identified by *n*, where n can be a value between 0 and 254. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_DISPARITY	Specifies that the component stores a disparity map. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_INFRARED	Specifies that the component stores an intensity image of infrared light.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_INTENSITY	Specifies that the component stores an intensity image of visible light.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_METADATA	Specifies that the component stores metadata information. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COMPONENT_MULTISPECTRAL	Specifies that the component stores an intensity image where each band represents the intensity of a specific wavelength of light. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_RANGE	Specifies that the component stores 3D distance/position information. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_REFLECTANCE	Specifies that the component stores a reflectance map. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_REGION_AIL	Specifies that the component stores a region of interest (ROI) for the M_COMPONENT_RANGE component of the container. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_SCATTER	Specifies that the component stores a scatter map. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_ULTRAVIOLET	Specifies that the component stores an intensity image of ultraviolet light.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_UNDEFINED	Specifies that the component stores information of an unknown type.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CUSTOM_DISTANCE_MAX		Sets the maximum difference within which two points are considered neighbors in the custom component. This control type is available only when you have specified a custom component with M_CUSTOM_DISTANCE_COMPONENT. If you set a non-infinite value, and the source container does not have the specified custom component, an Aurora Imaging Library error is reported. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INFINITE	Specifies that there is no custom component.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the maximum difference between points in the custom component.	c		e	f	g	a		i		j	k	l	m	n	b

M_GLOBAL_NORMAL_DISTANCE_MAX	Sets the maximum angle between the normal of a point and a blob's average normal, for the point to be added to the blob. This control type can be used to segment a point cloud into planar blobs to distinguish between relatively flat regions. You can specify how to calculate the angle with M_NORMAL_DISTANCE_MODE. Setting a smaller maximum angle forces the blobs into more planar (flatter) regions, allowing for less variation in each blob. Since this control is solely based upon normal vectors, parallel but distinct blobs can still merge together, unless M_GLOBAL_PLANE_DISTANCE_MAX is specified as well. This control type is available when an M_COMPONENT_NORMALS_AIL component exists in the source container. If the normals component is not present, the segmentation will fail and the operation's status returns M_MISSING_COMPONENT_NORMALS_AIL, when requested using M3dblobGetResult() with M_STATUS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies not to consider the angle.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 180.0	Specifies the maximum angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_PLANE_DISTANCE_MAX	Sets the maximum distance between a point and a blob's average plane, for the point to be added to the blob. The blob's average plane is defined by the average normal and passes through the centroid. This control type requires that M_GLOBAL_NORMAL_DISTANCE_MAX has been set, and is used to prevent parallel planar blobs from merging together. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies not to consider the distance.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_DISTANCE	Sets the maximum distance within which two points are considered neighbors. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_INFINITE	Specifies that there is no maximum distance. Instead, the function will use M_MAXIMUM_NUMBER_NEIGHBORS to determine a point's neighborhood.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the maximum distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_DISTANCE_MODE		Sets whether to use, for the maximum distance between points, the value set with M_MAX_DISTANCE or an automatic threshold. This is the maximum distance within which two points are considered neighbors. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_USER_DEFINED.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies to automatically calculate the maximum distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED	Specifies to use the value set with M_MAX_DISTANCE.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAXIMUM_NUMBER_NEIGHBORS		Sets the maximum number of neighbors to consider at each point. If there are more neighbors than the specified maximum, only the closest points are kept. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 49.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the maximum number of neighbor points.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEIGHBOR_SEARCH_MODE		Sets the search mode for finding the neighbors of a point. Note that M_ORGANIZED is usually faster, but is less robust to noise. M_TREE is typically the most precise because it chooses points that are closest in 3D space, using the Euclidean distance. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_TREE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MESH	Specifies to use the point cloud's mesh component to determine the neighbors of a point. A mesh's triangular faces are composed of connected neighboring points. > This option is available only when the source container has an M_COMPONENT_MESH_AIL component. If the mesh component is not present, the segmentation will fail and the operation's status returns M_MISSING_COMPONENT_MESH_AIL, when requested using M3dblobGetResult() with M_STATUS.	c		e	f	g	a		i		j	k	l	m	n	b

		M_ORGANIZED	Specifies to use the point cloud's organizational structure to determine the neighbors of a point. This option is supported only for an organized point cloud. When M_ORGANIZED is specified, you must set either M_MAXIMUM_NUMBER_NEIGHBORS or M_NEIGHBORHOOD_ORGANIZED_SIZE to determine the extent of the neighborhood. > If the point cloud is unorganized, the segmentation will fail and the operation's status returns M_CONTAINER_NOT_ORGANIZED, when requested using M3dblobGetResult() with M_STATUS.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TREE	Specifies to use a KD tree search mode to determine a point's neighbors. When M_TREE is specified, you must also set M_MAXIMUM_NUMBER_NEIGHBORS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NEIGHBORHOOD_ORGANIZED_SIZE		Sets the neighborhood size, which must be an odd number, when finding the neighbors of a point in an organized point cloud. The size determines the number of elements along a row or column of the 2D buffer of the range and confidence components. For example, if you set M_NEIGHBORHOOD_ORGANIZED_SIZE to 7, the neighborhood is a 7-element by 7-element region in the source container's 2D grid organization (specifically, in the organized point cloud's M_COMPONENT_RANGE and M_COMPONENT_CONFIDENCE components). Calculations for points close to the edge of the 2D buffer use a clipped neighborhood. This control type is supported only when using an organized point cloud and only if M_NEIGHBOR_SEARCH_MODE is set to M_ORGANIZED. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 7.	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 3	Specifies the organized neighborhood size, in number of pixels; the specified value must be an odd number.	c		e	f	g	a		i		j	k	l	m	n	b

M_NORMAL_DISTANCE_MAX	Sets the maximum angle between normal vectors within which two points are considered neighbors. Specify how to calculate the angle with M_NORMAL_DISTANCE_MODE. This control type is available when an M_COMPONENT_NORMALS_AIL component exists in the source container. If the normals component is not present, the segmentation will fail and the operation's status returns M_MISSING_COMPONENT_NORMALS_AIL, when requested using M3dblobGetResult() with M_STATUS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies not to consider the normals.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 180.0	Specifies the maximum angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_DISTANCE_MAX_MODE	Sets whether to use, for the maximum angle between normal vectors, the value set with M_NORMAL_DISTANCE_MAX or an automatic threshold. This is the maximum angle between the normal vectors of two neighboring points for the points to be considered neighbors. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_USER_DEFINED.	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically calculate the maximum angle.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED	Specifies to use the value set with M_NORMAL_DISTANCE_MAX.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_DISTANCE_MODE	Sets how to calculate the angle difference between the normal vectors of two points. If required, set M_NORMAL_DISTANCE_MODE to M_ORIENTATION to ensure that the segmentation is invariant to the normal viewpoint. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DIRECTION.	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION	Specifies to calculate the angle from 0 to 180 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_ORIENTATION	Specifies to calculate the angle from 0 to 90 degrees; when specified, this control value considers flipped normals to be the same.	c		e	f	g	a		i		j	k	l	m	n	b

M_NUMBER_OF_POINTS_MAX	Sets the maximum number of points for a blob to be considered valid. Blobs with more points than the specified value will be discarded and their points assigned the label 0. You can use M_NUMBER_OF_POINTS_MAX to remove the background, which is often the largest blob. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies that there is no maximum number of points.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the maximum number of points.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN	Sets the minimum number of points for a blob to be considered valid. Blobs with fewer points than the specified value will be discarded and their points will be assigned the label 0. You can use M_NUMBER_OF_POINTS_MIN to remove blobs corresponding to a few isolated points. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the minimum number of points.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELABEL_CONSECUTIVE	Sets whether to force the labels to be consecutive from 1 to M_MAX_LABEL_VALUE. When calling M3dblobSegment() with a label image, this will recompute the labels instead of using those from the label image. M_RELABEL_CONSECUTIVE sets each blob label to the blob's index + 1. Note that this operation will change the value of M_MAX_LABEL_VALUE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to force the labels to be consecutive.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to force the labels to be consecutive.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEGMENTATION_MODE	Sets the type of segmentation to perform. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_EUCLIDEAN.	c		e	f	g	a		i		j	k	l	m	n	b
M_EUCLIDEAN	Specifies to segment the point cloud based on the (Euclidean) distance between points. Use the other segmentation context control settings to modify the segmentation criteria.	c		e	f	g	a		i		j	k	l	m	n	b

	M_LABEL_IMAGE	Specifies to copy a label image into the specified result buffer (M3dblobSegment()). You can use M_LABEL_IMAGE to copy an external segmentation (for example, a label image created using the 2D Blob Analysis module). All segmentation control types except M_NUMBER_OF_POINTS_MAX, M_NUMBER_OF_POINTS_MIN, and M_RELABEL_CONSECUTIVE are ignored in this mode. Note that you can use M3dblobSegment() with the predefined context M_SEGMENTATION_CONTEXT_LABEL_IMAGE. In this case, all segmentation context control types are set to their default, except for M_SEGMENTATION_MODE which is set to M_LABEL_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WHOLE_IMAGE	Specifies to group all valid points into a single blob, which allows you to perform calculations on the entire point cloud. All segmentation control types except M_NUMBER_OF_POINTS_MAX and M_NUMBER_OF_POINTS_MIN are ignored in this mode. Note that you can use M3dblobSegment() with the predefined context M_SEGMENTATION_CONTEXT_WHOLE_IMAGE. In this case, all segmentation context control types are set to their default, except for M_SEGMENTATION_MODE which is set to M_WHOLE_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT		Sets the maximum amount of time for M3dblobSegment() to complete the blob segmentation operation before generating a time-out error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
Segmentation 3D blob analysis result ID, for a single blob or all blobs		Specifies a segmentation 3D blob analysis result buffer, allocated using M3dblobAllocResult() with M_SEGMENTATION_RESULT, and used to store M3dblobSegment() results.	c		e	f	g	a		i		j	k	l	m	n	b

	M_CUSTOM_FEATURE(AIL_INT N)	Adds or modifies the computed values of a custom feature of the blob(s). You can use a custom feature (using M3dblobSelect() and M3dblobSort()) as if it was a regular feature computed using M3dblobCalculate(). To add a custom feature, you can set M_CUSTOM_FEATURE() to hold a computation obtained from other feature results. For example, to add a measure of the blobs' semi-oriented box elongation (width divided by height), you can calculate the semi-oriented box (using M3dblobCalculate()), retrieve the box's width and height (using M3dblobGetResult()), and then use **M3dblobControl**() to add the width-divided-by-height value to M_CUSTOM_FEATURE(). The resulting custom feature is stored in the result buffer and is subsequently selectable and sortable using M3dblobSelect() and M3dblobSort(), respectively.	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters														
		N Specifies the index of the custom feature.														
		0 <= *Value* < 1024 Specifies the valid index of the custom feature.	c		e	f	g	a		i		j	k	l	m	n	b
	ControlValue	Specifies the custom feature's computed value.	c		e	f	g	a		i		j	k	l	m	n	b

M_MERGE	Merges the blob specified using the ControlValue parameter with the one specified using the LabelOrIndex parameter. The resulting blob retains the label of the LabelOrIndex blob. Since this operation changes the segmentation (unlike when using M3dblobSelect(), M3dblobSort(), and M3dblobCombine()), this result buffer becomes incompatible with sibling result buffers (for example, you cannot combine the set of blobs in this result buffer with those in sibling result buffers). Note that, since the merge changes the number of blobs, the index of any blob might change. If the two blobs contain calculated features, the merged blob will contain all features that can be trivially merged (for example, M_BOUNDING_BOX and M_PCA), but not features that would need to be recomputed from scratch (for example, minimum/maximum diameter and nearest blob). The merged blob will also never contain any custom features. For a successful merge, both blobs must be singular; that is, M_ALL_BLOBS must not be specified. If you want to treat all blobs as one, use M3dblobSegment() with either the predefined context M_SEGMENTATION_CONTEXT_WHOLE_IMAGE or a previously allocated M_SEGMENTATION_CONTEXT context for which you have set M_SEGMENTATION_MODE to M_WHOLE_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_INDEX(AIL_INT Index)	Specifies the index of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	Index Specifies the index.														
M_BLOB_LABEL(AIL_INT Label)	Specifies the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	Label Specifies the label.														
M_REMOVE_FEATURE	Sets the custom feature(s) to remove from the blob. If a feature is not available, no error is reported.	c		e	f	g	a		i		j	k	l	m	n	b
M_CUSTOM_FEATURE(AIL_INT N)	Removes a custom feature from the blob(s).	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	N Specifies the index of the custom feature.														

			0 <= *Value* < 1024 Specifies the valid index of the custom feature.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ALL	Specifies to remove all custom features.	c		e	f	g	a		i		j	k	l	m	n	b
Segmentation 3D blob analysis result ID, for general results			Specifies a segmentation 3D blob analysis result buffer, allocated using M3dblobAllocResult() with M_SEGMENTATION_RESULT, and used to store M3dblobSegment() results.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INDEX_MODE		Sets how to interpret the LabelOrIndex parameter in all functions (such as M3dblobGetResult()) that use the 3D blob analysis result buffer. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_USE_MACRO.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INDEX	Specifies that the index will be passed directly to the function's LabelOrIndex parameter, without any macros.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LABEL	Specifies that the label will be passed directly to the function's LabelOrIndex parameter, without any macros.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USE_MACRO	Specifies that the label or index must be passed using the macro M_BLOB_LABEL() or M_BLOB_INDEX().	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELABEL_CONSECUTIVE		Relabels the blobs in consecutive order, from 1 to M_MAX_LABEL_VALUE. M_RELABEL_CONSECUTIVE sets each blob label to the blob's index + 1. Note that this operation will change the value of M_MAX_LABEL_VALUE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RESET		Removes all results from the segmentation 3D blob analysis result buffer. This does not delete the result buffer, as is the case with M3dblobFree().	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b
M_STOP_SEGMENT		Stops the current execution of M3dblobSegment(). This can only be done from another thread of higher priority. Results already available in the result buffer become invalid and will be discarded.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for [when ContextOrResult3dblobId=Calculate 3D blob analysis context ID: M_FERET_GENERAL].

You can add one of the following values to the above-mentioned value to set the direction along which to calculate the custom Feret.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the custom Feret direction																		
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																	
M_FERET_DIRECTION_X		Sets the X-component of the direction vector.																
	Value	Specifies the X-component of the direction vector. The default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Y		Sets the Y-component of the direction vector.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-component of the direction vector. The default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Z		Sets the Z-component of the direction vector.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Z-component of the direction vector. The default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for [when ContextOrResult3dblobId=Calculate 3D blob analysis context ID: M_NEAREST_BLOB].

You can add the following value to the above-mentioned value to set the maximum distance to the nearest blob.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the maximum distance to the nearest blob																		
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																	
M_MAX_DISTANCE		Sets the maximum distance within which Aurora Imaging Library considers a nearby blob to be the closest.																

	M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no distance constraint.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the maximum distance. If the nearest blob is further than the specified value, the returned label will be 0.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void M3dblobControlDouble (AIL_ID ContextOrResult3dblobId, AIL_INT64 LabelOrIndex, AIL_INT64 ControlType,
AIL_DOUBLE ControlValue)
```

Details

Parameters

ContextOrResult3dblobId

See ContextOrResult3dblobId of the main function for a description.

LabelOrIndex

See LabelOrIndex of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void M3dblobControlInt64 (AIL_ID ContextOrResult3dblobId, AIL_INT64 LabelOrIndex, AIL_INT64 ControlType, AIL_INT64
ControlValue)
```

Details

Parameters

ContextOrResult3dblobId

See ContextOrResult3dblobId of the main function for a description.

LabelOrIndex

See LabelOrIndex of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.

DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobControlDraw

Synopsis

Control a setting of a draw 3D blob analysis context.

Syntax

```
void M3dblobControlDraw(
    AIL_ID DrawContext3dblobId, //out
    AIL_INT64 Operation,        //in
    AIL_INT64 ControlType,      //in
    AIL_DOUBLE ControlValue     //in
)
```

Description

This function controls a specified setting of a draw 3D blob analysis context. These settings establish which blob features to draw into the 3D graphics list (and how to draw them), when calling M3dblobDraw3d().

You can inquire about most of these settings using M3dblobInquireDraw().

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **M3dblobControlDraw()** internally calls the *AIL_DOUBLE* version of this function (M3dblobControlDrawDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (M3dblobControlDrawInt64()).

Parameters

DrawContext3dblobId

Specifies the identifier of the draw 3D blob analysis context to control. The 3D blob analysis context must have been previously allocated on the system using M3dblobAlloc() with M_DRAW_3D_CONTEXT.

Operation

Specifies the draw operation.

See the Parameter associations section for possible values that can be specified.

ControlType

Specifies the setting to change.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the setting's new value.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The table below lists possible values for the **Operation**, **ControlType**, and **ControlValue** parameters.

- For controlling a draw blob setting

The following Operation, ControlType, and ControlValue parameter settings can be specified for different types of features.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling a draw blob setting

Operation / ControlType / ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ALL	Applies the setting to all operations that support the control type.															
M_ACTIVE	Sets whether to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets the appearance of the graphic(s) to solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b

		M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
			Parameters															
			Red Specifies the red band, as a value between 0 and 255.															
			Green Specifies the green band, as a value between 0 and 255.															
			Blue Specifies the blue band, as a value between 0 and 255.															
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_BLOBS			Sets whether and how to draw the points inside each specified blob.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE		Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_PSEUDO_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_BOX	Sets whether and how to draw the bounding box of each specified blob. M_BOUNDING_BOX must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets the appearance of the graphic(s) to solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_WIREFRAME.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The 3D graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_WHITE.	c		e	f	g	a		i		j	k	l	m	n	b

		M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
			Parameters															
			Red Specifies the red band, as a value between 0 and 255.															
			Green Specifies the green band, as a value between 0 and 255.															
			Blue Specifies the blue band, as a value between 0 and 255.															
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_EXCLUDED_POINTS			Sets whether and how to draw the points that are not part of each specified blob (or not part of any blob).	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_GRAY.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b

M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FERET_GENERAL	Sets whether and how to draw the custom Feret of each specified blob. M_FERET_GENERAL and M_FERET_CONTACT_POINTS must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_WHITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FERET_MAX	Sets whether and how to draw the maximum diameter Feret of each specified blob. M_FERET_MAX_DIAMETER and M_FERET_CONTACT_POINTS must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_WHITE.	c		e	f	g	a		i		j	k	l	m	n	b

		M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
			Parameters															
			Red Specifies the red band, as a value between 0 and 255.															
			Green Specifies the green band, as a value between 0 and 255.															
			Blue Specifies the blue band, as a value between 0 and 255.															
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FERET_MIN			Sets whether and how to draw the minimum diameter Feret of each specified blob. M_FERET_MIN_DIAMETER and M_FERET_CONTACT_POINTS must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_WHITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_INDEX	Sets whether and how to draw the index at the centroid of each specified blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_YELLOW.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															

		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_COLOR	Specifies to draw the index for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_SIZE	Sets the size of the font with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the font size. This is the height of one line of text, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RENDER_LAYER	Sets the render layer on which to draw the graphic(s). The graphics on higher layers are always drawn completely in front of graphics on lower layers even if those on higher layers are farther away. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 2.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= 9	Specifies the layer on which the 3D graphic will be rendered.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_LABEL	Sets whether and how to draw the label at the centroid of each specified blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_YELLOW.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the label for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FONT_SIZE		Sets the size of the font with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the font size. This is the height of one line of text, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RENDER_LAYER	Sets the render layer on which to draw the graphic(s). The graphics on higher layers are always drawn completely in front of graphics on lower layers even if those on higher layers are farther away. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 2.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= 9	Specifies the layer on which the 3D graphic will be rendered.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_NEAREST_BLOB	Sets whether and how to draw the shortest line from each specified blob to its nearest blob. M_NEAREST_BLOB must have been calculated beforehand. If M_NEAREST_BLOB was calculated but there was no nearest blob, nothing is drawn.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_WHITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															

			Blue Specifies the blue band, as a value between 0 and 255.															
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b

		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PCA			Sets whether and how to draw the principal component of each specified blob; this is drawn as perpendicular axes centered on each blob's centroid and oriented along the axis directions determined from a principal component analysis (PCA). M_PCA must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE		Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PCA_BOX	Sets whether and how to draw the PCA-aligned bounding box of each specified blob. M_PCA_BOX must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets the appearance of the graphic(s) to solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_WIREFRAME.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b

M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_WHITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b

M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_SEMI_ORIENTED_BOX	Specifies whether and how to draw the semi-oriented bounding box of each specified blob. M_SEMI_ORIENTED_BOX must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets the appearance of the graphic(s) to solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_WIREFRAME.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The 3D graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Same as M_COLOR_WHITE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
			Parameters															
			Red Specifies the red band, as a value between 0 and 255.															
			Green Specifies the green band, as a value between 0 and 255.															
			Blue Specifies the blue band, as a value between 0 and 255.															
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_DRAW_SETTINGS			Sets a global draw 3D blob analysis context setting.	c		e	f	g	a		i		j	k	l	m	n	b

	M_PSEUDO_COLOR_OFFSET	Sets the offset to apply to the blob's label when drawing with M_PSEUDO_COLOR. When drawing blobs from different segmentations into the same display, Aurora Imaging Library assigns identical color sequences to the different segmentations, which causes the graphics for different blobs to receive the same color. If you increment the value assigned to M_PSEUDO_COLOR_OFFSET by the maximum label value of the blobs in the previously drawn segmentation (M3dblobGetResult() with M_MAX_LABEL_VALUE), the colors assigned to the current segmentation will be unique. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 0	Specifies the offset to apply to the label.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void M3dblobControlDrawDouble (AIL_ID DrawContext3dblobId, AIL_INT64 Operation, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

DrawContext3dblobId

See DrawContext3dblobId of the main function for a description.

Operation

See Operation of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void M3dblobControlDrawInt64 (AIL_ID DrawContext3dblobId, AIL_INT64 Operation, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

DrawContext3dblobId

See DrawContext3dblobId of the main function for a description.

Operation

See Operation of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobCopyResult

Synopsis

Copy a group of results into an image buffer, a 3D geometry object, a transformation matrix object, or a 3D blob analysis result buffer.

Syntax

```
void M3dblobCopyResult(
    AIL_ID SrcResult3dblobId,  //in
    AIL_INT64 SrcLabelOrIndex, //in
    AIL_ID DstAilObjectId,     //out
    AIL_INT64 CopyType,        //in
    AIL_INT64 ControlFlag      //in
)
```

Description

This function copies a group of 3D blob analysis results into an image buffer, a 3D geometry object, a transformation matrix object, or a 3D blob analysis result buffer, according to the specified copy operation.

Parameters

SrcResult3dblobId

Specifies the identifier of the 3D blob analysis result buffer from which to copy. The result buffer must have been previously allocated using M3dblobAllocResult() with M_SEGMENTATION_RESULT.

SrcLabelOrIndex

Specifies the label or index of the blob(s) from which to copy results, or specifies that you are copying general results for all blobs for which data is stored in the source result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the blob label or index																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_INDEX(AIL_INT Index)	Specifies the index of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Specifies the index.															

		0 <= *Value* < M_NUMBER Specifies the valid index of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_LABEL(AIL_INT Label)	Specifies the label of the blob.		c		e	f	g	a		i		j	k	l	m	n	b
	Parameters																
	Label Specifies the label.																
		1 <= *Value* <= M_MAX_LABEL_VALUE Specifies the valid label of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL_BLOBS	Specifies to copy results from all blobs.		c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to copy general results from the 3D blob analysis result buffer.		c		e	f	g	a		i		j	k	l	m	n	b

DstAilObjectId

Specifies the identifier of the image buffer, 3D geometry object, transformation matrix object, or 3D blob analysis result buffer in which to store the copied result.

See the Parameter associations section for possible values that can be returned.

CopyType

Specifies the type of copy operation to perform.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Parameter associations

The tables below list possible values for the **CopyType** parameter and possible values returned to the **DstAilObjectId** parameter.

- For copying general results from the source 3D blob analysis result buffer
- For copying the results of a specific blob or all blobs from the source 3D blob analysis result buffer
- For copying the results of a specific blob from the source 3D blob analysis result buffer

To copy general results, set the CopyType parameter to one of the following values. In this case, you must set SrcLabelOrIndex to M_GENERAL or M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For copying general results from the source 3D blob analysis result buffer

CopyType DstAilObjectId - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INDEX_IMAGE	Specifies to copy blob indices into an image buffer, creating an index image. For each point in a blob, this operation copies the blob's index to a corresponding pixel location in the image buffer. This essentially sets each pixel in the index image to the index of its corresponding blob. Note that the point information is set in the image buffer at the same location as the point's location in the range component of its point cloud container. Points that are not part of any blob are given the value of -1 in signed buffers, and set to the maximum buffer value in unsigned buffers.															
Image buffer identifier	Specifies the identifier of an image buffer in which to copy results. The image buffer must be a 1-band buffer and have a bit depth deep enough to hold the largest index + 1 (M_NUMBER). The image buffer size must equal the size of the container used for the segmentation, in both X and Y. To retrieve the size, use M3dblobGetResult() with M_RESULT_IMAGE_SIZE_X and M_RESULT_IMAGE_SIZE_Y. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_IMAGE	Specifies to copy blob labels into an image buffer, creating a label image. For each point in a blob, this operation copies the blob's label to a corresponding pixel location in the image buffer. This essentially sets each pixel in the label image to the label of its corresponding blob. Note that the point information is set in the image buffer at the same location as the point's location in the range component of its point cloud container. Points that are not part of any blob are given the label 0.	c		e	f	g	a		i		j	k	l	m	n	b

Image buffer identifier	Specifies the identifier of an image buffer in which to copy results. The image buffer must be a 1-band buffer and have a bit depth deep enough to hold the largest label (M_MAX_LABEL_VALUE). The image buffer size must equal the size of the container used for the segmentation, in both X and Y. To retrieve the size, use M3dblobGetResult() with M_RESULT_IMAGE_SIZE_X and M_RESULT_IMAGE_SIZE_Y. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT	Specifies to copy the whole source 3D blob analysis result buffer into the specified destination 3D blob analysis result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
Segmentation 3D blob analysis result buffer identifier	Specifies the identifier of the 3D blob analysis result buffer in which to copy results. The result buffer must have been previously allocated using M3dblobAllocResult() with M_SEGMENTATION_RESULT.	c		e	f	g	a		i		j	k	l	m	n	b

To copy the results of a specific blob or all blobs, set the CopyType parameter to the following value. In this case, you must set SrcLabelOrIndex to a specific blob, using M_BLOB_INDEX() or M_BLOB_LABEL(), or to M_ALL_BLOBS for all blobs.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For copying the results of a specific blob or all blobs from the source 3D blob analysis result buffer																
CopyType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
DstAilObjectId - Possible values returned																
M_MASK_IMAGE	Specifies to create a mask image whose pixels are non-zero for blob points, and zero otherwise. For each point in a blob, this operation uses the point's location in its container to assign a non-zero value to a corresponding pixel location in the image buffer. The non-zero value is 1 for floating-point image buffers and -1 for signed image buffers. For unsigned image buffers, the maximum buffer value is used. All other pixels in the buffer are set to 0.															

Image buffer identifier	Specifies the identifier of an image buffer in which to copy results. The image buffer must be a 1-band buffer and its size must equal the size of the container used for the segmentation, in both X and Y. To retrieve the size, use M3dblobGetResult() with M_RESULT_IMAGE_SIZE_X and M_RESULT_IMAGE_SIZE_Y. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b

To copy the results of a specific blob, set the CopyType parameter to one of the following values. In this case, you must set SrcLabelOrIndex to a specific blob using M_BLOB_INDEX() or M_BLOB_LABEL().

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For copying the results of a specific blob from the source 3D blob analysis result buffer																
CopyType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
DstAilObjectId - Possible values returned																
M_BOUNDING_BOX	Specifies to copy the blob's axis-aligned bounding box into the specified 3D geometry object, establishing a 3D box geometry.															
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER	Specifies to copy the center of the blob's axis-aligned bounding box into the specified 3D geometry object, establishing a 3D point geometry.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID	Specifies to copy the centroid of the blob into the specified 3D geometry object, establishing a 3D point geometry.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_CONTACT_POINT_1 +	Specifies to copy the start point of the blob's specified Feret diameter into the specified 3D geometry object, establishing a 3D point geometry. You must specify a combination value from the following table: • To specify the Feret from which to copy the contact point	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINT_2 +	Specifies to copy the end point of the blob's specified Feret diameter into the specified 3D geometry object, establishing a 3D point geometry. You must specify a combination value from the following table: • To specify the Feret from which to copy the contact point	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIXTURING_MATRIX	Specifies to copy, into the transformation matrix, the principal component analysis (PCA) results that can bring points along the blob's principal axes (around the centroid) to the X-, Y-, and Z-axes (around the origin). This matrix is the inverse of the PCA matrix (M_PCA_MATRIX).	c		e	f	g	a		i		j	k	l	m	n	b
Transformation matrix object ID in which to copy	Specifies the identifier of the transformation matrix object in which to copy the principal component analysis (PCA) results. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX.	c		e	f	g	a		i		j	k	l	m	n	b

M_NEAREST_BLOB +	Specifies to copy, into the specified 3D geometry object, the point inside the current blob which is closest to the nearest neighboring blob in the point cloud (or vice versa), establishing a 3D point geometry. You must add a combination constant to specify which of the points to copy, either the point in the current blob (M_NEAREST_POINT_1), or that of the nearest neighboring blob (M_NEAREST_POINT_2). You must specify a combination value from the following table: • To specify which of the closest points between the current blob and its nearest neighbor to copy	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_PCA_BOX	Specifies to copy the blob's PCA-aligned bounding box into the specified 3D geometry object, establishing a 3D box geometry.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_PCA_BOX_CENTER	Specifies to copy the center of the blob's PCA-aligned bounding box into the specified 3D geometry object, establishing a 3D point geometry.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_PCA_MATRIX	Specifies to copy, into the transformation matrix, the principal component analysis (PCA) results that can bring points on the X-, Y-, and Z-axes (around the origin) to the blob's principal axes (around the centroid).	c		e	f	g	a		i		j	k	l	m	n	b

Transformation matrix object ID in which to copy	Specifies the identifier of the transformation matrix object in which to copy the principal component analysis (PCA) results. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX.	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1	Specifies to copy the blob's first principal axis into the specified 3D geometry object, establishing a 3D line geometry that starts at the blob's centroid, and has a length of 1 unit.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2	Specifies to copy the blob's second principal axis into the specified 3D geometry object, establishing a 3D line geometry that starts at the blob's centroid, and has a length of 1 unit.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3	Specifies to copy the blob's third principal axis into the specified 3D geometry object, establishing a 3D line geometry that starts at the blob's centroid, and has a length of 1 unit.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX	Specifies to copy the blob's semi-oriented bounding box into the specified 3D geometry object, establishing a 3D box geometry.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b

M_SEMI_ORIENTED_BOX_CENTER	Specifies to copy the center of the blob's semi-oriented bounding box into the specified 3D geometry object, establishing a 3D point geometry.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object ID in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_NEAREST_BLOB.

You must add one of the following values to the above-mentioned value to specify which of the closest points between the current blob and its nearest neighbor to copy.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the point																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NEAREST_POINT_1	Specifies to copy the point, inside the current blob, which is closest to the nearest neighboring blob in the point cloud.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_2	Specifies to copy the point, inside the nearest neighboring blob in the point cloud, which is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_CONTACT_POINT_1; M_FERET_CONTACT_POINT_2.

You must add one of the following values to the above-mentioned values to specify the Feret from which to copy the contact point.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the Feret																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_AT_PRINCIPAL_AXIS_1	Specifies to copy the contact point of the Feret along the first principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_2	Specifies to copy the contact point of the Feret along the second principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_3	Specifies to copy the contact point of the Feret along the third principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL	Specifies to copy the contact point of the custom Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER	Specifies to copy the contact point of the longest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_MIN_DIAMETER	Specifies to copy the contact point of the shortest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_HEIGHT	Specifies to copy the contact point of the smallest XY-side of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_WIDTH	Specifies to copy the contact point of the longest XY-side of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_X	Specifies to copy the contact point of the X-aligned Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Y	Specifies to copy the contact point of the Y-aligned Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Z	Specifies to copy the contact point of the Z-aligned Feret.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobDraw3d

Synopsis

Draw blobs and blob features into a 3D graphics list.

Syntax

```
AIL_INT64 M3dblobDraw3d(
    AIL_ID OperationDraw3dContext3dblobId, //in
    AIL_ID ContainerBufId,                 //in
    AIL_ID Result3dblobId,                 //out
    AIL_INT64 LabelOrIndex,                //in
    AIL_ID DstList3dgraId,                 //in
    AIL_INT64 DstParentLabel,              //in
    AIL_INT64 ControlFlag                  //in
)
```

Description

This function draws blob points and/or features into the specified 3D graphics list. Set the draw operations and options for the draw using M3dblobControlDraw().

Note that the specified draw 3D blob analysis result buffer must contain the result of an M3dblobSegment() operation, and, if you want to draw blob features, results of an M3dblobCalculate() operation. Use M3dblobControlDraw() to set whether to draw blob points, excluded points, and/or calculated blob features.

The resulting draw has the following structure: Top node --> Blobs --> Features. The subnodes for blobs 0, 1, ...N are ordered in the 3D graphics list in the same sequence as the blob indices. If the excluded points are drawn (M_DRAW_EXCLUDED_POINTS set to M_ENABLE, using M3dblobControlDraw()), they are the last subnode.

Parameters

OperationDraw3dContext3dblobId

Specifies the identifier of the draw 3D blob analysis context that specifies the features to draw and how to draw them.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the draw 3D blob analysis context identifier																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies a predefined draw 3D blob analysis context with all draw operations (M3dblobControlDraw()) set to use default settings.	c		e	f	g	a		i		j	k	l	m	n	b
Draw 3D blob analysis context identifier	Specifies a valid draw 3D blob analysis context identifier, previously allocated using M3dblobAlloc() with M_DRAW_3D_CONTEXT.	c		e	f	g	a		i		j	k	l	m	n	b

ContainerBufId

Specifies the identifier of the 3D-processable point cloud container from which to retrieve the blobs to draw. This must be the same container previously passed to M3dblobSegment(), and its confidence component must not have been modified since then. If the segmentation was performed using a label image instead, this can be any container as long as the size of its range component is the same as the label image.

The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must be 3D-

processable (that is, if you call MbufInquireContainer() with M_3D_PROCESSABLE, the function returns M_PROCESSABLE).

Result3dblobId

Specifies the identifier of the 3D blob analysis result buffer from which to retrieve calculated results and/or information about the blobs. The result buffer must have been previously allocated using M3dblobAllocResult().

LabelOrIndex

Specifies the label or index of the blob whose points and/or calculated features you want to draw, or specifies all blobs. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the blob label or index																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_BLOB_INDEX(AIL_INT Index)	Specifies the index of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Specifies the index.															
	0 <= *Value* < M_NUMBER Specifies the valid index of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_LABEL(AIL_INT Label)	Specifies the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Label Specifies the label.															
	1 <= *Value* <= M_MAX_LABEL_VALUE Specifies the valid label of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL_BLOBS	Specifies to draw the points and/or features of all blobs.	c		e	f	g	a		i		j	k	l	m	n	b

DstList3dgraId

Specifies the identifier of the 3D graphics list in which to draw. You can specify a 3D graphics list that you have previously allocated using M3dgraAlloc(), or you can specify the identifier of the 3D display's internal graphics list (inquired using M3ddispInquire() with M_3D_GRAPHIC_LIST_ID).

DstParentLabel

Specifies the label of the 3D graphic in the 3D graphics list to use as the annotation's parent.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j –

radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the parent label																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ROOT_NODE.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROOT_NODE	Specifies the top-most node of the 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the label of the 3D graphic in the 3D graphics list. Label 0 is the 3D graphics list's root node.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Return value

Returns the parent label of all 3D graphics that the function added to the 3D graphics list. If the specified 3D blob analysis result buffer does not contain a blob segmentation, a node 3D graphic is added to the graphics list and the node's label is returned.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobExtract

Synopsis

Populate a container with the specified blob(s).

Syntax

```
void M3dblobExtract(
    AIL_ID SrcContainerBufId, //in
    AIL_ID Result3dblobId,    //out
    AIL_INT64 LabelOrIndex,   //in
    AIL_ID DstContainerBufId, //in
    AIL_INT64 Options,        //in
    AIL_INT64 ControlFlag     //in
)
```

Description

This function copies the specified blob(s) from the source to the destination container. This is useful when you want to calculate a feature or perform an operation that is too complex for M3dblobCalculate(), such as fitting a geometry to a specific blob (M3dmetFit()), or creating a mesh for a specific blob (M3dimMesh()).

> Note that this function never copies an M_COMPONENT_MESH_AIL component to the destination container, even if you specify to extract all blobs.

Parameters

SrcContainerBufId

Specifies the identifier of the 3D-processable point cloud container from which to extract the blobs. This must be the same container previously passed to M3dblobSegment(), and its confidence component must not have been modified since then. If the segmentation was performed using a label image instead, this can be any container as long as the size of the container's range component is the same as the label image.

The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must be 3D-processable (that is, if you call MbufInquireContainer() with M_3D_PROCESSABLE, the function returns M_PROCESSABLE).

Result3dblobId

Specifies the identifier of the 3D blob analysis result buffer in which the blob segmentation is stored. The result buffer must have been previously allocated using M3dblobAllocResult().

LabelOrIndex

Specifies the label or index of the blob to extract, or specifies to extract all blobs. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the blob label or index																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_BLOB_INDEX(AIL_INT Index)	Specifies the index of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

	Index Specifies the index.															
	0 <= *Value* < M_NUMBER Specifies the valid index of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_LABEL(AIL_INT Label)	Specifies the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Label Specifies the label.															
	1 <= *Value* <= M_MAX_LABEL_VALUE Specifies the valid label of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL_BLOBS	Specifies to extract all blobs.	c		e	f	g	a		i		j	k	l	m	n	b

DstContainerBufId

Specifies the identifier of the destination point cloud container in which to save the extracted blobs.

The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must not be a child container.

Options

Specifies how to organize the destination point cloud.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying how to organize the destination point cloud																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_AUTO	Chooses the fastest option. This depends on the size and density of the blob(s) relative to the source container.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAME	Specifies to copy all the source container's components (except any mesh component) into the destination container; no change is made to the size of the components. > The confidence score for non-extracted blob points is set to zero.	c		e	f	g	a		i		j	k	l	m	n	b

M_SHRINK	Specifies to copy, from source to destination, the smallest region that holds all valid points of the extracted blob(s), in all components (except any mesh component). Note that the region is a rectangular region within the bands of the 2D data buffers that hold point cloud data. > The confidence score for non-extracted blob points is set to zero.	c		e	f	g	a		i		j	k	l	m	n	b
M_SHRINK_VERTICALLY	Specifies to copy all the source container's components (except any mesh component) into the destination container, maintaining the source's size in X (that is, the width of its range component), but shrinking the size in Y to the smallest interval that contains all extracted blob points. You can use M_SHRINK_VERTICALLY to merge consecutive scans in continuous scanning applications. > The confidence score for non-extracted blob points is set to zero.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNORGANIZED	Specifies to store results in an unorganized point cloud. Components in the destination container will store results of the extract operation in 1D, unorganized buffers, ordered in the same manner as M_PIXEL_X and M_PIXEL_Y.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Specifies which components will be affected by the operation.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether to apply the operation to all components																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the operation will only affect the core components of a 3D-processable container. These components are listed in the description of MbufInquireContainer() with M_3D_PROCESSABLE. All other components are copied to the destination container unmodified.	c		e	f	g	a		i		j	k	l	m	n	b

M_APPLY_TO_ALL_COMPONENTS	Specifies that the operation will affect the core components of a 3D-processable container, as well as any other component (including custom components) that has the same dimension as M_COMPONENT_RANGE. These components must not have a region of interest (ROI) associated with them. For example, using a component that is an image buffer with an ROI will cause an error. > Note that if a custom component has an associated calibration context, the component's information is extracted, but its calibration context is discarded. If the resulting operation yields an empty container with 0 valid points, the components with the same dimensions are not created and will not exist in the destination container.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobFree

Synopsis

Free a 3D blob analysis context or a 3D blob analysis result buffer.

Syntax

```
void M3dblobFree(
    AIL_ID ContextOrResult3dblobId //in
)
```

Description

This function deletes the specified 3D blob analysis context or 3D blob analysis result buffer, and releases any memory associated with it.

All 3D blob analysis contexts and 3D blob analysis result buffers allocated on a particular system must be freed before the system can be freed, unless M_UNIQUE_ID was specified during allocation.

Parameter

ContextOrResult3dblobId

Specifies the identifier of the 3D blob analysis context or 3D blob analysis result buffer to free. These must have been successfully allocated (using M3dblobAlloc() or M3dblobAllocResult()) prior to calling this function.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobGetResult

Synopsis

Get calculated feature results for the specified blobs, from a 3D blob analysis result buffer.

Syntax

```
AIL_DOUBLE M3dblobGetResult(
    AIL_ID Result3dblobId,  //in
    AIL_INT64 LabelOrIndex, //in
    AIL_INT64 ResultType,   //in
    void *ResultArrayPtr    //out
)
```

Description

This function retrieves the results for a specified feature of one or all blobs, from the specified 3D blob analysis result buffer.

The specified feature(s) must have already been calculated using M3dblobCalculate(). Note that **M3dblobGetResult**() generates an error if you try to retrieve a result whose corresponding feature was not calculated. You can add the M_AVAILABLE combination constant to a feature to determine if the result is available for retrieval.

Parameters

Result3dblobId

Specifies the identifier of the 3D blob analysis result buffer from which to get results.

LabelOrIndex

Specifies the label or index of the blob for which to get results, or specifies that you are getting general results related to all blobs. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the blob label or index																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_INDEX(AIL_INT Index)	Specifies the index of the blob listed in the specified result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Specifies the index.															
	0 <= *Value* < M_NUMBER Specifies the valid index of a blob.	c		e	f	g	a		i		j	k	l	m	n	b

M_BLOB_LABEL(AIL_INT Label)	Specifies the label of the blob listed in the specified result buffer. Note that blob labels are not consecutive; use M_INDEX_VALUE to establish if a blob with a given label exists. With the exception of M_INDEX_VALUE, if you specify a blob label that does not exist, Aurora Imaging Library returns an error.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	Label Specifies the label.														
	1 <= *Value* <= M_MAX_LABEL_VALUE Specifies the valid label of a blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL_BLOBS	Specifies to retrieve results from all blobs listed in the specified result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to retrieve general results from the 3D blob analysis result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result to retrieve.

See the Parameter associations section for possible values that can be specified.

ResultArrayPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]
- array of type AIL_FLOAT [optionally, in C++: a reference to a std::vector<AIL_FLOAT>]
- array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]
- array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]
- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]
- AIL_DOUBLE
- AIL_FLOAT
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the results. Since the **M3dblobGetResult**() function also returns the results, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **ResultType** parameter and possible values returned to the **ResultArrayPtr** parameter.

- For retrieving general results from the 3D blob analysis result buffer
- For retrieving results that are always available from a single blob or all blobs

- For retrieving results that are sometimes available from a single blob or all blobs

To retrieve a general result, the ResultType parameter should be set to one of the following values. In this case, you must set LabelOrIndex to M_GENERAL or M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving general results from the 3D blob analysis result buffer																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_CONTAINER_ID +	Retrieves the identifier of the container that was used for the segmentation. If M3dblobSegment() has not been called yet or if it was called with a label image instead of a container, M_NULL is returned. If this result is not null, post-segmentation functions that take a container (such as M3dblobCalculate() and M3dblobExtract()) expect this container as input.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies that M3dblobSegment() has not been called or that it was called with a label image instead of a container.	c		e	f	g	a		i		j	k	l	m	n	b
Container identifier	Specifies the Aurora Imaging Library identifier of the container that was used for the segmentation.	c		e	f	g	a		i		j	k	l	m	n	b
M_INDEX_IMAGE_SIZE_BIT +	Retrieves the minimum bit size required for the index image, assuming an unsigned buffer. Note that you can use this value to allocate an image buffer of an appropriate size for M3dblobCopyResult() to create an index image.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_IMAGE_SIZE_BIT +	Retrieves the minimum bit size required for the label image, assuming an unsigned buffer. Note that you can use this value to allocate an image buffer of an appropriate size for M3dblobCopyResult() to create a label image.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

M_MAX_DISTANCE +	Retrieves the maximum distance within which two points are considered neighbors, depending on the maximum distance mode. If M_MAX_DISTANCE_MODE was not set to M_USER_DEFINED, this is the value that was computed during segmentation.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no maximum distance.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the maximum distance value.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_LABEL_VALUE +	Retrieves the maximum label value of all blobs in the result buffer. Label values for blobs are generated when a call to M3dblobCalculate() is made. The maximum label value is not necessarily the same as the total number of blobs (M_NUMBER) because some label values might not be used. You can call **M3dblobGetResult**() with M_LABEL_IMAGE_SIZE_BIT to determine the bit depth required when creating a label image using M3dblobCopyResult(). Note that if zero is returned, no blobs were found.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_DISTANCE_MAX +	Retrieves the maximum angle between normal vectors within which two points are considered neighbors. If M_NORMAL_DISTANCE_MAX_MODE was not set to M_USER_DEFINED, this is the value that was computed during segmentation.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no maximum angle.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the maximum angle value.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER +	Retrieves the number of blobs that are listed in the result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

M_RESULT_IMAGE_SIZE_X +	Retrieves the size in X of the container or image used for the blob segmentation. Note that you can use this value to allocate an image buffer of an appropriate size for M3dblobCopyResult() to create an index image, label image, or mask image.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the size in X, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT_IMAGE_SIZE_Y +	Retrieves the size in Y of the container or image used for the blob segmentation. Note that you can use this value to allocate an image buffer of an appropriate size for M3dblobCopyResult() to create an index image, label image, or mask image.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the size in Y, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_STATUS +	Retrieves the status of the segmentation operation, or why the segmentation failed.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTAINER_NOT_ORGANIZED	Specifies that the segmentation failed because the source point cloud container was unorganized.	c		e	f	g	a		i		j	k	l	m	n	b
M_MISSING_COMPONENT_MESH_AIL	Specifies that the segmentation failed because the source point cloud container did not have an M_COMPONENT_MESH_AIL component.	c		e	f	g	a		i		j	k	l	m	n	b
M_MISSING_COMPONENT_NORMALS_AIL	Specifies that the segmentation failed because the source point cloud container did not have an M_COMPONENT_NORMALS_AIL component.	c		e	f	g	a		i		j	k	l	m	n	b
M_NOT_INITIALIZED	Specifies that the 3D blob analysis result buffer was not used in a call to M3dblobSegment(), and contains no results.	c		e	f	g	a		i		j	k	l	m	n	b

M_STOPPED_BY_REQUEST	Specifies that the blob segmentation operation was stopped from another thread using M3dblobControl() with M_STOP_SEGMENT.	c		e	f	g	a		i		j	k	l	m	n	b
M_SUCCESS	Specifies that the blob segmentation operation completed successfully.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT_REACHED	Specifies that the blob segmentation operation took longer than the allowed value, specified using M3dblobControl() with M_TIMEOUT, and has stopped before completion.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOTAL_NUMBER_OF_POINTS +	Retrieves the total number of points across all blobs.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve an always-available result of an M3dblobSegment() operation, the ResultType parameter should be set to one of the following values. In this case, you must set LabelOrIndex to M_BLOB_INDEX(), M_BLOB_LABEL(), or M_ALL_BLOBS. If LabelOrIndex is M_ALL_BLOBS, the results are returned in an array instead of a single value.

> These results are available after an M3dblobSegment() operation, and remain available after an M3dblobCalculate() operation that uses the results of the blob segmentation. If M3dblobSegment() has not yet been called on this result, you will receive an error.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving results that are always available from a single blob or all blobs																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_BLOB_TOUCHING_IMAGE_BORDERS +	Retrieves information about which side(s) of the container or label image that the blob touches. This is either the side(s) of an organized point cloud container's range component or a 2D label image. Note that M_BLOB_TOUCHING_IMAGE_BORDERS is available only when the object used for segmentation (M3dblobSegment() with ContainerOrLabelImageBufId) is an organized point cloud or 2D label image. Note that a combination of the values below can be returned. For example, if the blob touches the bottom-left, the function returns M_BOTTOM + M_LEFT. Bitwise operators must be used to verify the borders that the blob touches.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b

	M_NULL	Specifies that no blob touches a side.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOTTOM	Specifies that the blob touches the bottom of the container or label image.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LEFT	Specifies that the blob touches the left of the container or label image.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RIGHT	Specifies that the blob touches the right of the container or label image.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TOP	Specifies that the blob touches the top of the container or label image.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_X +		Retrieves the X-coordinate of the first pixel in the blob. This is the left-most pixel on the top-most line of the blob's stored location in the container's 2D grid organization, or it is the equivalent pixel in the label image. The value returned is the same as the first X-coordinate returned with M_PIXEL_X or M_PIXEL_PACKED.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_Y +		Retrieves the Y-coordinate of the first pixel in the blob. This is the left-most pixel on the top-most line of the blob's stored location in the container's 2D grid organization, or it is the equivalent pixel in the label image. The value returned is the same as the first Y-coordinate returned with M_PIXEL_Y or M_PIXEL_PACKED.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_INDEX_VALUE +		Retrieves the index of the blob. You can use this result type to establish if a blob with a given label exists. If the label is valid, M_INDEX_VALUE returns the corresponding index; if the label is not valid, M_INVALID is returned instead of an Aurora Imaging Library error.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INVALID	Specifies an invalid index.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* < M_NUMBER	Specifies the index of the blob.	c		e	f	g	a		i		j	k	l	m	n	b

M_LABEL_VALUE +	Retrieves the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
0 < *Value* <= M_MAX_LABEL_VALUE	Specifies the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS +	Retrieves the number of points in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_X +	Retrieves the maximum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_Y +	Retrieves the maximum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_X +	Retrieves the minimum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_Y +	Retrieves the minimum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_PIXEL_PACKED +	Retrieves the packed X- and Y-coordinates of the blob's stored location in the container's 2D grid organization (or the equivalent coordinates in the label image). The coordinates are returned in an interleaved manner: [X_0, Y_0, X_1, Y_1, ...]. Note that the order is from left to right, top to bottom, starting with the blob's top-left stored location in the container (or equivalent location in the label image).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_X +	Retrieves the X-coordinates of the blob's stored location in the container's 2D grid organization (or the equivalent coordinates in the label image).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_Y +	Retrieves the Y-coordinates of the blob's stored location in the container's 2D grid organization (or the equivalent coordinates in the label image).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve a sometimes-available result of an M3dblobSegment() or M3dblobCalculate() operation, the ResultType parameter should be set to one of the following values. In this case, you must set LabelOrIndex to M_BLOB_INDEX(), M_BLOB_LABEL(), or M_ALL_BLOBS. If LabelOrIndex is M_ALL_BLOBS, the results are returned in an array instead of a single value.

> If M3dblobSegment() has not yet been called on this result, an error is thrown.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving results that are sometimes available from a single blob or all blobs																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_CUSTOM_FEATURE(AIL_INT N) +	Retrieves the computed values of the specified custom feature that was manually added to the source result buffer using M3dblobControl() with M_CUSTOM_FEATURE(). This feature was not calculated using M3dblobCalculate().															
	Parameters															
	N Specifies the index of the custom feature.															

		0 <= *Value* < 1024 Specifies the valid index of the custom feature.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*		c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_CENTRAL_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ) +	Retrieves the central moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where μ is the centroid and (x_i , y_i , z_i) is the i th point: $\sum_i (x_i - \mu_x)^{PowerX} (y_i - \mu_y)^{PowerY} (z_i - \mu_z)^{PowerZ}$ To retrieve this result, you must have enabled M_MOMENTS (using M3dblobControl()) and set M_MOMENT_ORDER to an order that is greater than or equal to PowerX + PowerY + PowerZ.		c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PowerX Specifies the power in X.															
		Value >= 0 Specifies the power.														
	PowerY Specifies the power in Y.															
		Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
	PowerZ Specifies the power in Z.															
		Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*		c		e	f	g	a		i		j	k	l	m	n	b

M_MOMENT_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ) +	Retrieves the (ordinary, non-central) moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where (x_i, y_i, z_i) is the ith point: $\sum_i x_i^{PowerX} y_i^{PowerY} z_i^{PowerZ}$ To retrieve this result, you must have enabled M_MOMENTS (using M3dblobControl()) and set M_MOMENT_ORDER to an order that is greater than or equal to PowerX + PowerY + PowerZ.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	PowerX Specifies the power in X.														
	Value >= 0 — Specifies the power.														
	PowerY Specifies the power in Y.														
	Value >= 0 — Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
	PowerZ Specifies the power in Z.														
	Value >= 0 — Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_X +	Retrieves the X-component of the blob's average normal vector. To retrieve this result, you must have set M_GLOBAL_NORMAL_DISTANCE_MAX (using M3dblobControl()) to a specific angle value (not M_INFINITE). This result is internally calculated during the segmentation, provided the above condition.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_AVERAGE_NORMAL_Y +	Retrieves the Y-component of the blob's average normal vector. To retrieve this result, you must have set M_GLOBAL_NORMAL_DISTANCE_MAX (using M3dblobControl()) to a specific angle value (not M_INFINITE). This result is internally calculated during the segmentation, provided the above condition.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_Z +	Retrieves the Z-component of the blob's average normal vector. To retrieve this result, you must have set M_GLOBAL_NORMAL_DISTANCE_MAX (using M3dblobControl()) to a specific angle value (not M_INFINITE). This result is internally calculated during the segmentation, provided the above condition.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_X +	Retrieves the X-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX. To retrieve this result, you must have enabled M_BOUNDING_BOX, M_PCA_BOX, or M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_Y +	Retrieves the Y-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX. To retrieve this result, you must have enabled M_BOUNDING_BOX, M_PCA_BOX, or M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_BOX_CENTER_Z +	Retrieves the Z-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX. To retrieve this result, you must have enabled M_BOUNDING_BOX, M_PCA_BOX, or M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_X +	Retrieves the X-coordinate of the centroid of the blob. To retrieve this result, you must have enabled M_CENTROID (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 1.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Y +	Retrieves the Y-coordinate of the centroid of the blob. To retrieve this result, you must have enabled M_CENTROID (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 1.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Z +	Retrieves the Z-coordinate of the centroid of the blob. To retrieve this result, you must have enabled M_CENTROID (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 1.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_COVARIANCE_MATRIX +	Retrieves the 9 entries that make up the covariance matrix, in row-major order. The covariance matrix is defined as a 3x3 matrix of second order central moments: xx xy xz xy yy yz xz yz zz To retrieve this result, you must have enabled M_MOMENTS (using M3dblobControl()) and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_1 +	Retrieves the eigenvalue corresponding to the first principal axis. The first principal axis is the direction along which there is the most variance. To retrieve this result, you must have enabled M_MOMENTS (using M3dblobControl()) and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_2 +	Retrieves the eigenvalue corresponding to the second principal axis. The second principal axis is the direction along which there is the second most variance, perpendicular to the first principal axis. To retrieve this result, you must have enabled M_MOMENTS (using M3dblobControl()) and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_3 +	Retrieves the eigenvalue corresponding to the third principal axis. The third principal axis is perpendicular to the first and second principal axes. To retrieve this result, you must have enabled M_MOMENTS (using M3dblobControl()) and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_AT_PRINCIPAL_AXIS_1 +	Retrieves the length of the Feret along the first principal axis. This result is the same as M_SIZE_X + M_PCA_BOX. To retrieve this result, you must have enabled M_PCA_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_2 +	Retrieves the length of the Feret along the second principal axis. This result is the same as M_SIZE_Y + M_PCA_BOX. To retrieve this result, you must have enabled M_PCA_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_3 +	Retrieves the length of the Feret along the third principal axis. This result is the same as M_SIZE_Z + M_PCA_BOX. To retrieve this result, you must have enabled M_PCA_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL +	Retrieves the length of the Feret in the direction specified with M_FERET_GENERAL + M_FERET_DIRECTION_.... To retrieve this result, you must have enabled M_FERET_GENERAL (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER +	Retrieves the length of the longest Feret in the blob. Note that, when you combine this result type with an M_FERET_DIRECTION_... combination value, the direction is accurate up to the angle specified with M_FERET_PRECISION (using M3dblobControl()). To retrieve this result, you must have enabled M_FERET_MAX_DIAMETER (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_MIN_DIAMETER +	Retrieves the length of the shortest Feret in the blob. Note that, when you combine this result type with an M_FERET_DIRECTION_... combination value, the direction is accurate up the angle specified with M_FERET_PRECISION (using M3dblobControl()). To retrieve this result, you must have enabled M_FERET_MIN_DIAMETER (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_HEIGHT +	Retrieves the height (smallest XY side) of the semi-oriented box. This result is the same as M_SIZE_Y + M_SEMI_ORIENTED_BOX. To retrieve this result, you must have enabled M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_WIDTH +	Retrieves the width (longest XY side) of the semi-oriented box. This result is the same as M_SIZE_X + M_SEMI_ORIENTED_BOX. To retrieve this result, you must have enabled M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_X +	Retrieves the length of the X-aligned Feret (same as M_SIZE_X). To retrieve this result, you must have enabled M_BOUNDING_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Y +	Retrieves the length of the Y-aligned Feret (same as M_SIZE_Y). To retrieve this result, you must have enabled M_BOUNDING_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b

ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Z +	Retrieves the length of the Z-aligned Feret (same as M_SIZE_Z). To retrieve this result, you must have enabled M_BOUNDING_BOX or M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_LINEARITY +	Retrieves the degree to which the specified blob is linear, where 1.0 is a perfect line and 0.0 is a perfect sphere. This is equal to the following equation: 1 - sqrt(M_EIGENVALUE_2/M_EIGENVALUE_1). To retrieve this result, you must have enabled M_LINEARITY (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 1.0	Specifies the degree to which the blob is linear.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_X +	Retrieves the X-coordinate of the maximum corner of the blob's axis-aligned bounding box. To retrieve this result, you must have enabled M_BOUNDING_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Y +	Retrieves the Y-coordinate of the maximum corner of the blob's axis-aligned bounding box. To retrieve this result, you must have enabled M_BOUNDING_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_MAX_Z +	Retrieves the Z-coordinate of the maximum corner of the blob's axis-aligned or semi-oriented bounding box. To retrieve this result, you must have enabled M_BOUNDING_BOX or M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_X +	Retrieves the X-coordinate of the minimum corner of the blob's axis-aligned bounding box. To retrieve this result, you must have enabled M_BOUNDING_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Y +	Retrieves the Y-coordinate of the minimum corner of the blob's axis-aligned bounding box. To retrieve this result, you must have enabled M_BOUNDING_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Z +	Retrieves the Z-coordinate of the minimum corner of the blob's axis-aligned or semi-oriented bounding box. To retrieve this result, you must have enabled M_BOUNDING_BOX or M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB +	Retrieves the label of the nearest blob. If there are no other blobs, or if the nearest blob is further than M_NEAREST_BLOB + M_MAX_DISTANCE, the label is 0 and no other nearest blob results are available. To retrieve this result, you must have enabled M_NEAREST_BLOB (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_NEAREST_BLOB_DISTANCE +	Retrieves the minimum distance to the nearest blob. To retrieve this result, you must have enabled M_NEAREST_BLOB (using M3dblobControl()) and there must be at least one blob within the specified maximum distance (that is, **M3dblobGetResult**() with M_NEAREST_BLOB must not return 0).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANARITY +	Retrieves the degree to which the specified blob is planar, where 1.0 is a perfect plane and 0.0 is a perfect sphere. This is equal to the following equation: 1 - sqrt(M_EIGENVALUE_3/ M_EIGENVALUE_2). To retrieve this result, you must have enabled M_PLANARITY (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 1.0	Specifies the degree to which the blob is planar.	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_X +	Retrieves the X-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. To retrieve this result, you must have enabled M_PCA (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_Y +	Retrieves the Y-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. To retrieve this result, you must have enabled M_PCA (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_PRINCIPAL_AXIS_1_Z +	Retrieves the Z-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. To retrieve this result, you must have enabled M_PCA (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_X +	Retrieves the X-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. To retrieve this result, you must have enabled M_PCA (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_Y +	Retrieves the Y-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. To retrieve this result, you must have enabled M_PCA (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_Z +	Retrieves the Z-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. To retrieve this result, you must have enabled M_PCA (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_PRINCIPAL_AXIS_3_X +	Retrieves the X-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. To retrieve this result, you must have enabled M_PCA (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Y +	Retrieves the Y-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. To retrieve this result, you must have enabled M_PCA (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Z +	Retrieves the Z-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. To retrieve this result, you must have enabled M_PCA (using M3dblobControl()). Otherwise, you must have enabled M_MOMENTS and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX_ANGLE +	Retrieves the angle of the semi-oriented bounding box's rotation around the Z-axis, measured from the X-axis to the box's longest XY-side, respecting the right-hand rule. To retrieve this result, you must have enabled M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 180.0	Specifies the angle of the semi-oriented bounding box.	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_X +	Retrieves the size in X of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX. To retrieve this result, you must have enabled M_BOUNDING_BOX, M_PCA_BOX, or M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +	Retrieves the size in Y of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX. To retrieve this result, you must have enabled M_BOUNDING_BOX, M_PCA_BOX, or M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z +	Retrieves the size in Z of the 3D blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX. To retrieve this result, you must have enabled M_BOUNDING_BOX, M_PCA_BOX, or M_SEMI_ORIENTED_BOX (using M3dblobControl()).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_X +	Retrieves the standard deviation of points in the blob along the X-axis. This is equivalent to the following formula, where N is the number of points in the blob, and x_i and μ_x are the X-coordinates of the blob's i th point and centroid, respectively: $\sqrt{\frac{1}{N-1}\sum_i (x_i - \mu_x)^2}$ To retrieve this result, you must have enabled M_MOMENTS (using M3dblobControl()) and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b

ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Y +	Retrieves the standard deviation of points in the blob along the Y-axis. This is equivalent to the following formula, where *N* is the number of points in the blob, and y_i and μ_y are the Y-coordinates of the blob's i^{th} point and centroid, respectively: $\sqrt{\frac{1}{N-1}\sum_i (y_i - \mu_y)^2}$ To retrieve this result, you must have enabled M_MOMENTS (using M3dblobControl()) and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Z +	Retrieves the standard deviation of points in the blob along the Z-axis. This is equivalent to the following formula, where *N* is the number of points in the blob, and z_i and μ_z are the Z-coordinates of the blob's i^{th} point and centroid, respectively: $\sqrt{\frac{1}{N-1}\sum_i (z_i - \mu_z)^2}$ To retrieve this result, you must have enabled M_MOMENTS (using M3dblobControl()) and set M_MOMENT_ORDER to an order of at least 2.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_BOX_CENTER_X; M_BOX_CENTER_Y; M_BOX_CENTER_Z; M_SIZE_X; M_SIZE_Y; M_SIZE_Z.

You can add one of the following values to the above-mentioned values to specify the type of bounding box for which to return results.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of bounding box																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_PCA_BOX	Specifies to return results associated with the PCA bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX	Specifies to return results associated with the semi-oriented bounding box.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER;

M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to retrieve the contact points of the Feret.

To retrieve these results, you must have enabled M_FERET_CONTACT_POINTS (using M3dblobControl()).

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving Feret contact points																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_CONTACT_POINTS_X1	Retrieves the X-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_X2	Retrieves the X-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y1	Retrieves the Y-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y2	Retrieves the Y-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z1	Retrieves the Z-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z2	Retrieves the Z-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER; M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to retrieve the direction of the Feret.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the Feret direction																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_DIRECTION_X	Retrieves the X-component of the Feret's direction vector. For example, the direction of M_FERET_X is (1, 0, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Y	Retrieves the Y-component of the Feret's direction vector. For example, the direction of M_FERET_Y is (0, 1, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Z	Retrieves the Z-component of the Feret's direction vector. For example, the direction of M_FERET_Z is (0, 0, 1).	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_NEAREST_BLOB.

You can add one of the following values to the above-mentioned value to get the coordinates of the closest points between the current blob and its nearest neighbor.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the closest points between neighboring 3D blobs																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NEAREST_POINT_X1	Retrieves the X-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_X2	Retrieves the X-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Y1	Retrieves the Y-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Y2	Retrieves the Y-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z1	Retrieves the Z-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z2	Retrieves the Z-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for

Tables:

- For retrieving general results from the 3D blob analysis result buffer
- For retrieving results that are always available from a single blob or all blobs
- For retrieving results that are sometimes available from a single blob or all blobs

the values listed in all the parameter association tables.

You can add the following value to the above-mentioned values to determine whether a result is available.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining whether results are available																	
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ResultArrayPtr - Possible values returned																
M_AVAILABLE		Retrieves whether the requested result type is available for retrieval.															
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the requested result type is not available.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the requested result type is available.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for

Tables:

- For retrieving general results from the 3D blob analysis result buffer
- For retrieving results that are always available from a single blob or all blobs
- For retrieving results that are sometimes available from a single blob or all blobs

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested information to the required data type.

> Note: In C++, when using a standard vector (std::vector) instead of an array, the values listed in this combination table must not be used. If you need to use a specific data type, instantiate the vector with the necessary data type.

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For specifying the data type																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_FLOAT	Casts the requested information to an *AIL_FLOAT*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_FLOAT* **|** *array of type AIL_FLOAT [optionally, in C++: a reference to a std::vector<AIL_FLOAT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT32* **|** *array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64* **|** *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_DOUBLE*. If the requested information does not fit into an *AIL_DOUBLE*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobInquire

Synopsis

Inquire about a 3D blob analysis context or result buffer setting.

Syntax

```
AIL_INT64 M3dblobInquire(
    AIL_ID ContextOrResult3dblobId, //in
    AIL_INT64 LabelOrIndex,         //in
    AIL_INT64 InquireType,          //in
    void *UserVarPtr                //out
)
```

Description

This function inquires about a specified setting of a 3D blob analysis context or result buffer.

To inquire about draw 3D blob analysis settings for M3dblobDraw3d(), use M3dblobInquireDraw() instead.

Parameters

ContextOrResult3dblobId

Specifies the identifier of the 3D blob analysis context or result buffer to inquire.

See the Parameter associations section for possible values that can be specified.

LabelOrIndex

Specifies that you are inquiring about a global setting of the 3D blob analysis context or result buffer.

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For specifying what to inquire																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default value. If a 3D blob analysis context is specified, same as M_CONTEXT. If a 3D blob analysis result buffer is specified, same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT	Specifies to inquire a general setting of a 3D blob analysis context.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to inquire a general setting of a 3D blob analysis result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

InquireType

Specifies the type of setting about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_FLOAT
- AIL_INT

- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **M3dblobInquire**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **ContextOrResult3dblobId** and **InquireType** parameters and possible values returned to the **UserVarPtr** parameter.

- For inquiring a 3D blob analysis context
- For inquiring a 3D blob analysis result buffer

For the following inquire types, the ContextOrResult3dblobId parameter must specify a 3D blob analysis context.

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For inquiring a 3D blob analysis context																	
ContextOrResult3dblobId InquireType UserVarPtr - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Calculate 3D blob analysis context ID	Specifies a calculate 3D blob analysis context. allocated using M3dblobAlloc() with M_CALCULATE_CONTEXT, and used in M3dblobCalculate() operations. In this case, you must set LabelOrIndex to M_CONTEXT or M_DEFAULT.																
M_BOUNDING_BOX +	Inquires whether to calculate the axis-aligned bounding box and its related Ferets (M_FERET_...). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the axis-aligned bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the axis-aligned bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID +	Inquires whether to calculate the centroid. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

M_DISABLE	Specifies not to calculate the centroid.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the centroid.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS +	Inquires whether to save the contact points at each Feret; the contact points are the coordinates of the Feret diameter's start and end points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to save the contact points at each Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to save the contact points at each Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL +	Inquires whether to calculate the Feret along a custom direction specified with M_FERET_GENERAL + M_FERET_DIRECTION_.... SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the Feret along a custom direction.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the Feret along a custom direction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER +	Inquires whether to calculate the longest Feret. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to calculate the longest Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the longest Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_DIAMETER +	Inquires whether to calculate the shortest Feret. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies not to calculate the shortest Feret.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to calculate the shortest Feret.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_PRECISION +		Inquires the accuracy when calculating the minimum and maximum Feret directions. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* < 90.0	Specifies the precision (in degrees).	c		e	f	g	a		i		j	k	l	m	n	b
M_LINEARITY +		Inquires whether to calculate the degree to which the blobs are linear. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to calculate the linearity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to calculate the linearity.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_ORDER +		Inquires the order up to which moments are calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the moment order.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENTS +		Inquires whether to calculate all moments up to M_MOMENT_ORDER. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to calculate the moments.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to calculate the moments.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB +		Inquires whether to calculate features related to the nearest blob. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies not to calculate features related to the nearest blob.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to calculate features related to the nearest blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_PCA +		Inquires whether to calculate the principal components, which are used in a principal component analysis (PCA) or when calculating M_PCA_BOX. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to calculate the principal components.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to calculate the principal components.	c		e	f	g	a		i		j	k	l	m	n	b
M_PCA_BOX +		Inquires whether to calculate the PCA-aligned bounding box and its related Ferets. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to calculate the PCA-aligned bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to calculate the PCA-aligned bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANARITY +		Inquires whether to calculate the degree to which the blobs are planar. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to calculate the planarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to calculate the planarity.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX +		Inquires whether to calculate the semi-oriented bounding box and its related Ferets. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

M_DISABLE	Specifies not to calculate the semi-oriented bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to calculate the semi-oriented bounding box and its related Ferets.	c		e	f	g	a		i		j	k	l	m	n	b
Segmentation 3D blob analysis context ID	Specifies a segmentation 3D blob analysis context. allocated using M3dblobAlloc() with M_SEGMENTATION_CONTEXT, and used in M3dblobSegment() operations. In this case, you must set LabelOrIndex to M_CONTEXT or M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DISTANCE_MAX +	Inquires the maximum difference between the color of two points for them to be considered neighbors. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies not to consider the color.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum color difference.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DISTANCE_MODE +	Inquires how to calculate the difference in color between two points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_EUCLIDEAN	Specifies to use the Euclidean distance to calculate the color difference. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MANHATTAN	Specifies to use the Manhattan distance to calculate the color difference. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CUSTOM_DISTANCE_COMPONENT +	Inquires whether to use an extra component during the segmentation. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

		M_NULL	Specifies that there is no custom component.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_CONFIDENCE	Specifies that the component stores confidence information for the M_COMPONENT_RANGE or M_COMPONENT_DISPARITY component of the container. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_COORDINATE_MAP_A_AIL	Specifies that the component stores the A coordinate map provided by the camera. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_COORDINATE_MAP_B_AIL	Specifies that the component stores the B coordinate map provided by the camera. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_CUSTOM + *n*	Specifies that the component has a custom component type, identified by *n*, where n can be a value between 0 and 254. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_DISPARITY	Specifies that the component stores a disparity map. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_INFRARED	Specifies that the component stores an intensity image of infrared light. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_INTENSITY	Specifies that the component stores an intensity image of visible light. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_METADATA	Specifies that the component stores metadata information. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_MULTISPECTRAL	Specifies that the component stores an intensity image where each band represents the intensity of a specific wavelength of light. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_RANGE	Specifies that the component stores 3D distance/position information. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_REFLECTANCE	Specifies that the component stores a reflectance map. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPONENT_REGION_AIL	Specifies that the component stores a region of interest (ROI) for the M_COMPONENT_RANGE component of the container. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_COMPONENT_SCATTER	Specifies that the component stores a scatter map. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COMPONENT_ULTRAVIOLET	Specifies that the component stores an intensity image of ultraviolet light. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COMPONENT_UNDEFINED	Specifies that the component stores information of an unknown type. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Bit-encoded value that represents the component	Specifies an encoded value that represents the component type, or, if a macro was used, the macro used and the component's index or Aurora Imaging Library identifier. You can use the **M_COMPONENT_EXTRACT_VALUE** macro to retrieve the component type, or, if a macro was used, the component's index or Aurora Imaging Library identifier. To retrieve the type of macro that was used to pass the component, use the **M_COMPONENT_EXTRACT_FLAG** macro, which returns **M_COMPONENT_BY_ID_FLAG** or **M_COMPONENT_BY_INDEX_FLAG**; 0 is returned if no macro was used. Note that the encoded value can be passed wherever the component can be expressed using its type or using a macro (for example, Component parameter of MbufInquireContainer()).	c		e	f	g	a		i		j	k	l	m	n	b
M_CUSTOM_DISTANCE_MAX +		Inquires the maximum difference within which two points are considered neighbors in the custom component. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies that there is no custom component.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the maximum difference between points in the custom component.	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_NORMAL_DISTANCE_MAX +		Inquires the maximum angle between the normal of a point and a blob's average normal, for the point to be added to the blob. SET	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_INFINITE	Specifies not to consider the angle.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 < *Value* <= 180.0	Specifies the maximum angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GLOBAL_PLANE_DISTANCE_MAX +		Inquires the maximum distance between a point and a blob's average plane, for the point to be added to the blob. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_INFINITE	Specifies not to consider the distance.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the maximum distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_DISTANCE +		Inquires the maximum distance within which two points are considered neighbors. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_INFINITE	Specifies that there is no maximum distance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the maximum distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_DISTANCE_MODE +		Inquires whether to use, for the maximum distance between points, the value set with M_MAX_DISTANCE or an automatic threshold. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AUTO	Specifies to automatically calculate the maximum distance.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the value set with M_MAX_DISTANCE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAXIMUM_NUMBER_NEIGHBORS +		Inquires the maximum number of neighbors to consider at each point. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

Value >= 0	Specifies the maximum number of neighbor points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NEIGHBOR_SEARCH_MODE +	Inquires the search mode for finding the neighbors of a point. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_MESH	Specifies to use the point cloud's mesh component to determine the neighbors of a point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ORGANIZED	Specifies to use the point cloud's organizational structure to determine the neighbors of a point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TREE	Specifies to use a KD tree search mode to determine a point's neighbors. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NEIGHBORHOOD_ORGANIZED_SIZE +	Inquires the neighborhood size, when finding the neighbors of a point in an organized point cloud. This inquire type is supported only when using an organized point cloud and only if M_NEIGHBOR_SEARCH_MODE is set to M_ORGANIZED. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 3	Specifies the organized neighborhood size, in number of pixels; the specified value must be an odd number. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_DISTANCE_MAX +	Inquires the maximum angle between normal vectors within which two points are considered neighbors. This inquire type is available when an M_COMPONENT_NORMALS_AIL component exists in the source container. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies not to consider the normals.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 180.0	Specifies the maximum angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_NORMAL_DISTANCE_MAX_MODE +	Inquires whether to use, for the maximum angle between normal vectors, the value set with M_NORMAL_DISTANCE_MAX or an automatic threshold. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically calculate the maximum angle.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED	Specifies to use the value set with M_NORMAL_DISTANCE_MAX.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_DISTANCE_MODE +	Inquires how to calculate the angle difference between the normal vectors of two points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION	Specifies to calculate the angle from 0 to 180 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_ORIENTATION	Specifies to calculate the angle from 0 to 90 degrees; when specified, this control value considers flipped normals to be the same.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MAX +	Inquires the maximum number of points for a blob to be considered valid. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies that there is no maximum number of points.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the maximum number of points.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN +	Inquires the minimum number of points for a blob to be considered valid. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the minimum number of points.	c		e	f	g	a		i		j	k	l	m	n	b

M_RELABEL_CONSECUTIVE +	Inquires whether to force the labels to be consecutive from 1 to M_MAX_LABEL_VALUE. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to force the labels to be consecutive.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to force the labels to be consecutive.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEGMENTATION_MODE +	Inquires the type of segmentation to perform. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_EUCLIDEAN	Specifies to segment the point cloud based on the (Euclidean) distance between points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_IMAGE	Specifies to copy a label image into the specified result buffer (M3dblobSegment()). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WHOLE_IMAGE	Specifies to group all valid points into a single blob, which allows you to perform calculations on the entire point cloud. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT +	Inquires the maximum amount of time for M3dblobSegment() to complete the blob segmentation operation before generating a time-out error. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b

For the following inquire type, the ContextOrResult3dblobId parameter must specify a 3D blob analysis result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a 3D blob analysis result buffer																
ContextOrResult3dblobId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType																

		UserVarPtr - Possible values returned															
Segmentation 3D blob analysis result ID, for general results			Specifies a segmentation 3D blob analysis result buffer, allocated using M3dblobAllocResult() with M_SEGMENTATION_RESULT, and used to store M3dblobSegment() results.														
	M_INDEX_MODE +		Inquires how to interpret the LabelOrIndex parameter in all functions (including this function and, for example, M3dblobGetResult()) that use the 3D blob analysis result buffer. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_INDEX	Specifies that the index will be passed directly to the function's LabelOrIndex parameter, without any macros.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LABEL	Specifies that the label will be passed directly to the function's LabelOrIndex parameter, without any macros.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USE_MACRO	Specifies that the values passed to the LabelOrIndex parameter of the function can be the macros M_BLOB_INDEX() or M_BLOB_LABEL().	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for [when ContextOrResult3dblobId=Calculate 3D blob analysis context ID: M_NEAREST_BLOB].

You can add the following value to the above-mentioned value to get the maximum distance to the nearest blob.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the maximum distance to the nearest blob			(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType		Description															
	UserVarPtr - Possible values returned																
M_MAX_DISTANCE		Inquires the maximum distance within which Aurora Imaging Library considers a nearby blob to be the closest.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for [when ContextOrResult3dblobId=Calculate 3D blob analysis context ID: M_FERET_GENERAL].

You can add one of the following values to the above-mentioned value to get the direction along which to calculate M_FERET_GENERAL.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the Feret direction			(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType		Description															

UserVarPtr - Possible values returned																
M_FERET_DIRECTION_X	Inquires the X-component of the direction vector.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Y	Inquires the Y-component of the direction vector.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Z	Inquires the Z-component of the direction vector.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the InquireType parameter listed in For inquiring a 3D blob analysis context, For inquiring a 3D blob analysis result buffer.

You can add one of the following values to the above-mentioned values to get information about the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the default value																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_DEFAULT	Inquires the default value of the specified inquire type.															
UserVarPtr extra info	Data type info: Data type: *defined elsewhere*	c		e	f	g	a		i		j	k	l	m	n	b
M_IS_SET_TO_DEFAULT	Inquires whether the specified inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the inquire type is not set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the InquireType parameter listed in For inquiring a 3D blob analysis context, For inquiring a 3D blob analysis result buffer.

You can add one of the following values to the above-mentioned values to determine whether an inquire type has a default value or whether it is supported.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring whether an inquire type has a default value or whether it is supported																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

UserVarPtr - Possible values returned																
M_HAS_DEFAULT	Inquires whether the specified inquire type has a default value.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the inquire type does not have a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the inquire type has a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_SUPPORTED	Inquires whether the specified inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the InquireType parameter listed in For inquiring a 3D blob analysis context, For inquiring a 3D blob analysis result buffer.

You can add one of the following values to the above-mentioned values to cast the requested information to the required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_FLOAT	Casts the requested information to an *AIL_FLOAT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_FLOAT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT64*. If the requested information does not fit into an *AIL_INT64*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobInquireDraw

Synopsis

Inquire about a setting of a draw 3D analysis context.

Syntax

```
AIL_INT64 M3dblobInquireDraw(
    AIL_ID DrawContext3dblobId, //in
    AIL_INT64 Operation,        //in
    AIL_INT64 InquireType,      //in
    void *UserVarPtr            //out
)
```

Description

This function inquires about a specified setting of a draw 3D blob analysis context. Draw blob settings establish which blob features to draw into the 3D graphics list when calling M3dblobDraw3d() and how to draw them.

You can control the draw blob settings using M3dblobControlDraw().

Parameters

DrawContext3dblobId

Specifies the identifier of the draw 3D blob analysis context to inquire.

Operation

Specifies the draw operation about which to inquire.

See the Parameter associations section for possible values that can be specified.

InquireType

Specifies the type of setting about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_FLOAT
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **M3dblobInquireDraw**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The table below lists possible values for the **Operation** and **InquireType** parameters and possible values returned to the **UserVarPtr** parameter.

- For inquiring about a draw blob setting

The following Operation and InquireType parameter settings are available to inquire about draw blob settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a draw blob setting

Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType																
UserVarPtr - Possible values returned																
M_DRAW_BLOBS	Inquires whether and how to draw the points inside each specified blob.															
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_BOX			Inquires whether and how to draw the bounding box of each specified blob. M_BOUNDING_BOX must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s). This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
	M_APPEARANCE +		Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_WIREFRAME	Specifies a wireframe appearance. INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_EXCLUDED_POINTS	Inquires whether and how to draw the points that are not part of each specified blob (or not part of any blob).	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b

M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FERET_GENERAL	Inquires whether and how to draw the custom Feret of each specified blob. M_FERET_GENERAL and M_FERET_CONTACT_POINTS must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b

M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FERET_MAX	Inquires whether and how to draw the maximum diameter Feret of each specified blob. M_FERET_MAX_DIAMETER and M_FERET_CONTACT_POINTS must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FERET_MIN		Inquires whether and how to draw the minimum diameter Feret of each specified blob. M_FERET_MIN_DIAMETER and M_FERET_CONTACT_POINTS must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

		Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_INDEX			Inquires whether and how to draw the index at the centroid of each specified blob.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the index for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_SIZE +		Inquires the size of the font with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the font size. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RENDER_LAYER +		Inquires the render layer on which to draw the graphic(s). The graphics on higher layers are always drawn completely in front of graphics on lower layers even if those on higher layers are farther away. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* <= 9	Specifies the layer on which the 3D graphic will be rendered. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_LABEL		Inquires whether and how to draw the label(s) at the centroid of each specified blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

		M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_YELLOW	Specifies the color yellow. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the label for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_SIZE +		Inquires the size of the font with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the font size. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RENDER_LAYER +		Inquires the render layer on which to draw the graphic(s). The graphics on higher layers are always drawn completely in front of graphics on lower layers even if those on higher layers are farther away. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* <= 9	Specifies the layer on which the 3D graphic will be rendered. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_NEAREST_BLOB	Inquires whether and how to draw the shortest line from each specified blob to its nearest blob. M_NEAREST_BLOB must have been calculated beforehand. If M_NEAREST_BLOB was calculated but there was no nearest blob, nothing is drawn.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_PCA	Inquires whether and how to draw the principal component of each specified blob; this is drawn as perpendicular axes centered on each blob's centroid and oriented along the axis directions determined from a principal component analysis (PCA). M_PCA must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PCA_BOX	Inquires whether and how to draw the PCA-aligned bounding box of each specified blob. M_PCA_BOX must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_WIREFRAME	Specifies a wireframe appearance. INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_WHITE	Specifies the color white. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_SEMI_ORIENTED_BOX	Specifies whether and how to draw the semi-oriented bounding box of each specified blob. M_SEMI_ORIENTED_BOX must have been calculated beforehand.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the graphic for each blob with a different color, according to the mapping between each blob's label value and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_DRAW_SETTINGS	Inquires a global draw 3D blob analysis context setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR_OFFSET +	Inquires the offset to apply to the blob's label when drawing with M_PSEUDO_COLOR. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the offset to apply to the label. The default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the InquireType parameter listed in For inquiring about a draw blob setting.

You can add one of the following values to the above-mentioned values to get information about the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the default value																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_DEFAULT	Inquires the default value of the specified inquire type.															
UserVarPtr extra info	Data type info: Data type: *defined elsewhere*	c		e	f	g	a		i		j	k	l	m	n	b
M_IS_SET_TO_DEFAULT	Inquires whether the specified inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the inquire type is not set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRUE	Specifies that the inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the InquireType parameter listed in For inquiring about a draw blob setting.

You can add one of the following values to the above-mentioned values to determine whether an inquire type has a default value or whether it is supported.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring whether an inquire type has a default value or whether it is supported																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_HAS_DEFAULT		Inquires whether the specified inquire type has a default value.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type does not have a default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type has a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_SUPPORTED		Inquires whether the specified inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the InquireType parameter listed in For inquiring about a draw blob setting.

You can add one of the following values to the above-mentioned values to cast the requested information to the required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE		Casts the requested information to an *AIL_DOUBLE*.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_FLOAT		Casts the requested information to an *AIL_FLOAT*.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_FLOAT*	c		e	f	g	a		i		j	k	l	m	n	b

M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT64*. If the requested information does not fit into an *AIL_INT64*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobRestore

Synopsis

Restores a 3D blob analysis context from disk.

Syntax

```
AIL_ID M3dblobRestore(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID SysId,                //in
    AIL_INT64 ControlFlag,       //in
    AIL_ID *Context3dblobIdPtr   //out
)
```

Description

This function restores a 3D blob analysis context that was previously saved to a file, using M3dblobSave() or M3dblobStream(). This function restores all of the 3D blob analysis context's settings that were in effect when it was saved.

When the restored 3D blob analysis context is no longer required, release it using M3dblobFree() unless M_UNIQUE_ID was specified during restoration; if M_UNIQUE_ID was specified, the smart identifier manages the 3D blob analysis context's lifetime and you must not manually free it.

Parameters

FileName

Specifies the name and path of the file from which to restore the 3D blob analysis context. The function handles (internally) the opening and closing of the file.

This parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Open** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). Typically, 3D blob analysis context files have an M3DBLOB extension. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

SysId

Specifies the system on which to restore the 3D blob analysis context.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j –

radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Context3dblobIdPtr

Specifies the address of the variable in which to write the 3D blob analysis context identifier or specifies the data type that the function should use to return the 3D blob analysis context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D blob analysis context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D blob analysis context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DBLOB_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D blob analysis context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b

Address in which to write the calculate 3D blob analysis context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated calculate 3D blob analysis context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the draw 3D blob analysis context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated draw 3D blob analysis context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the segmentation 3D blob analysis context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated segmentation 3D blob analysis context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 3D blob analysis context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DBLOB_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobSave

Synopsis

Save a 3D blob analysis context to a file.

Syntax

```
void M3dblobSave(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID Context3dblobId,      //in
    AIL_INT64 ControlFlag        //in
)
```

Description

This function saves all the information about a previously allocated 3D blob analysis context to disk.

To load a saved context, use either M3dblobRestore() or M3dblobStream().

Parameters

FileName

Specifies the name and path of the file in which to save the 3D blob analysis context; it is recommended that you use the M3DBLOB file extension for easier use with other Aurora Imaging software products. The function internally handles the opening and closing of this file. If this file already exists, it will be overwritten.

This parameter can be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Save As** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") [1]	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). The recommended extension is M3DBLOB. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Context3dblobId

Specifies the identifier of the 3D blob analysis context to save. The context must have been successfully allocated (using M3dblobAlloc()) prior to calling this function.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobSegment

Synopsis

Calculate and store a blob segmentation from a point cloud or label image.

Syntax

```
void M3dblobSegment(
    AIL_ID SegmentationContext3dblobId, //in
    AIL_ID ContainerOrLabelImageBufId,  //in
    AIL_ID Result3dblobId,              //out
    AIL_INT64 ControlFlag               //in
)
```

Description

This function calculates a blob segmentation and stores it in a 3D blob analysis result buffer. The blob segmentation is calculated from the specified point cloud or it is defined from the specified label image.

When you specify a point cloud, the blob segmentation is performed according to the settings of the segmentation 3D blob analysis context. To set up the context, use M3dblobControl().

When you specify a label image to define the blob segmentation, the label image is copied to the result buffer. For example, you can specify a label image created using the (2D) Blob Analysis module (specifically, a blob identifier image created using MblobLabel()). Each pixel in the label image is set to a value that will identify corresponding points in a point cloud as belonging to a specific blob, using a label that is unique for each blob. Pixels with a value of zero represent points that are not part of any blob. Note that the location of each pixel in the label image corresponds to the point at the same location in the point cloud container. Also note that invalid points in the point cloud must have a label of zero; otherwise, functions that use the points might try to operate on the invalid points, causing an error.

The blob segmentation defines which points are considered to be in the same blob when M3dblobCalculate() is called.

The result buffer will hold the blob segmentation as well as a list of blob labels, to which indices are assigned (in whatever order the blobs happen to be in the list). You can specify to relabel the blobs in consecutive order, using M3dblobControl() with M_RELABEL_CONSECUTIVE, for either a context (when calling **M3dblobSegment**()), or a segmentation result (after calling **M3dblobSegment**()).

Parameters

SegmentationContext3dblobId

Specifies a segmentation 3D blob analysis context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the segmentation 3D blob analysis context identifier																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_SEGMENTATION_CONTEXT_LABEL_IMAGE	Specifies a predefined segmentation 3D blob analysis context with all segmentation context control types (M3dblobControl()) set to their default, except for M_SEGMENTATION_MODE which is set to M_LABEL_IMAGE. Use this predefined context to copy an external label image into the 3D blob analysis result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

M_SEGMENTATION_CONTEXT_WHOLE_IMAGE	Specifies a predefined segmentation 3D blob analysis context with all segmentation context control types (M3dblobControl()) set to their default, except for M_SEGMENTATION_MODE which is set to M_WHOLE_IMAGE. Use this predefined context to count all valid points as belonging to a single blob; any point whose confidence score is 0 will not be included as part of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
Segmentation 3D blob analysis context identifier	Specifies the identifier of a segmentation 3D blob analysis context, previously allocated using M3dblobAlloc() with M_SEGMENTATION_CONTEXT. > If a previously allocated context is specified, the function applies the segmentation control settings specified using M3dblobControl().	c		e	f	g	a		i		j	k	l	m	n	b

ContainerOrLabelImageBufId

Specifies the identifier of the 3D-processable point cloud container or image buffer from which to calculate the segmentation. You can specify either a container or a label image, depending on the segmentation context's M_SEGMENTATION_MODE.

When M_SEGMENTATION_MODE is set to M_WHOLE_IMAGE, M_EUCLIDEAN, or M_DEFAULT, ContainerOrLabelImageBufId must be a container.

When M_SEGMENTATION_MODE is set to M_LABEL_IMAGE, ContainerOrLabelImageBufId must be an image in which each pixel is set to a value that will identify corresponding points in a point cloud as belonging to a specific blob, using a label that is unique for each blob. Pixels with a value of zero represent points that are not part of any blob.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the container or label image buffer identifier																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Label image buffer identifier	Specifies the identifier of a label image buffer. The label image must be a 1-band 8, 16, or 32-bit unsigned image buffer. If you specify an image buffer that has an ROI associated with it, the ROI must be in raster format; otherwise, you will get an error. Points outside the ROI will not be part of any blob.	c		e	f	g	a		i		j	k	l	m	n	b
Point cloud container identifier	Specifies the identifier of a point cloud container. The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must be 3D-processable (that is, if you call MbufInquireContainer() with M_3D_PROCESSABLE, the function returns M_PROCESSABLE). Note that the container must not be a child container on a Distributed Aurora Imaging Library remote system with a parent that contains multiple range or confidence components; otherwise, an error is generated.	c		e	f	g	a		i		j	k	l	m	n	b

Result3dblobId

Specifies the identifier of the segmentation 3D blob analysis result buffer. The result buffer must have been allocated using M3dblobAllocResult() with M_SEGMENTATION_RESULT.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobSelect

Synopsis

Select which blobs meet a criterion and set the destination result buffer to recognize only these as blobs; corresponding blob identifying information and feature results are copied to the specified destination result buffer.

Syntax

```
void M3dblobSelect(
    AIL_ID SrcResult3dblobId,     //in
    AIL_ID DstResult3dblobId,     //out
    AIL_INT64 SelectionCriterion, //in
    AIL_INT64 Condition,          //in
    AIL_DOUBLE CondLow,           //in
    AIL_DOUBLE CondHigh,          //in
    AIL_INT64 ControlFlag         //in
)
```

Description

This function selects which blobs meet a criterion and sets the destination result buffer to recognize only these as blobs; corresponding blob identifying information and feature results are copied to the specified destination result buffer.

To perform a compound selection (criterion1 AND/OR criterion2), call this function twice using two separate destination result buffers. Select criterion1 with the first call, and then select criterion2 with the second call. Then, pass the two result buffers to M3dblobCombine() to combine the selections into a single result buffer that holds only the required blobs' data (based on the specified M3dblobCombine() operation).

This function supports in-place processing; you can specify the identifier of the same buffer for the source and the destination. Note that non-selected blob identifying information and feature results are deleted when using in-place processing.

Parameters

SrcResult3dblobId

Specifies the identifier of the source 3D blob analysis result buffer from which to select blob data. This is the result buffer that stores the blob segmentation and any calculated feature results.

DstResult3dblobId

Specifies the identifier of the destination 3D blob analysis result buffer into which to copy the selected blob data. In-place processing is supported; DstResult3dblobId can refer to the same result buffer as SrcResult3dblobId.

SelectionCriterion

Specifies the feature upon which to base the selection. This can be any of the following features; these features correspond to any non-array feature available using M3dblobGetResult() that is also retrievable using a specific blob label.

Note that you can also select for feature results that are not available. To do so, specify the SelectionCriterion, and then set the Condition parameter to M_NOT_AVAILABLE.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the feature																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_CUSTOM_FEATURE(AIL_INT N)	Uses the computed values of the specified custom feature that was manually added to the source result buffer using M3dblobControl() with M_CUSTOM_FEATURE(). This feature was not calculated using M3dblobCalculate().	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	N Specifies the index of the custom feature.															
	0 <= *Value* < 1024 — Specifies the valid index of the custom feature.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_CENTRAL_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ)	Uses the central moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where *N* is the number of points in the blob, μ is the centroid, and (x_i, y_i, z_i) is the ith point: $\sum_i (x_i - \mu_x)^{PowerX} (y_i - \mu_y)^{PowerY} (z_i - \mu_z)^{PowerZ}$	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PowerX Specifies the power in X.															
	Value >= 0 — Specifies the power.															
	PowerY Specifies the power in Y.															
	Value >= 0 — Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
	PowerZ Specifies the power in Z.															
	Value >= 0 — Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b

M_MOMENT_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ)	Uses the (ordinary, non-central) moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where *N* is the number of points in the blob and (x_i, y_i, z_i) is the ith point: $\sum_i x_i^{PowerX} y_i^{PowerY} z_i^{PowerZ}$	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	PowerX Specifies the power in X.														
	Value >= 0 — Specifies the power.														
	PowerY Specifies the power in Y.														
	Value >= 0 — Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
	PowerZ Specifies the power in Z.														
	Value >= 0 — Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_X	Uses the X-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_Y	Uses the Y-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_Z	Uses the Z-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_TOUCHING_IMAGE_BORDERS	Uses information about which side(s) of the container or label image that the blob touches. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_X +	Uses the X-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b

M_BOX_CENTER_Y +	Uses the Y-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_Z +	Uses the Z-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_X	Uses the X-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Y	Uses the Y-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Z	Uses the Z-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_1	Uses the eigenvalue corresponding to the first principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_2	Uses the eigenvalue corresponding to the second principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_3	Uses the eigenvalue corresponding to the third principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_1 +	Uses the length of the Feret along the first principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_2 +	Uses the length of the Feret along the second principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_3 +	Uses the length of the Feret along the third principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL +	Uses the length of the Feret in the custom direction that was specified using M3dblobControl() with M_FERET_GENERAL + M_FERET_DIRECTION_....	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER +	Uses the length of the longest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_DIAMETER +	Uses the length of the shortest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_SEMI_ORIENTED_HEIGHT +	Uses the height (smallest XY side) of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_WIDTH +	Uses the width (longest XY side) of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_X +	Uses the length of the X-aligned Feret (same as M_SIZE_X).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Y +	Uses the length of the Y-aligned Feret (same as M_SIZE_Y).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Z +	Uses the length of the Z-aligned Feret (same as M_SIZE_Z).	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_X	Uses the X-coordinate of the first pixel in the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_Y	Uses the Y-coordinate of the first pixel in the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_INDEX_VALUE	Uses the index of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_VALUE	Uses the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINEARITY	Uses the degree to which the specified blob is linear, where 1.0 is a perfect line and 0.0 is a perfect sphere. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_X	Uses the X-coordinate of the maximum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Y	Uses the Y-coordinate of the maximum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Z	Uses the Z-coordinate of the maximum corner of the blob's axis-aligned or semi-oriented bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_X	Uses the X-coordinate of the minimum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Y	Uses the Y-coordinate of the minimum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Z	Uses the Z-coordinate of the minimum corner of the blob's axis-aligned or semi-oriented bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_NEAREST_BLOB +	Uses the label of the nearest blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB_DISTANCE	Uses the minimum distance to the nearest blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS	Uses the number of points in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_X	Uses the maximum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_Y	Uses the maximum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_X	Uses the minimum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_Y	Uses the minimum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANARITY	Uses the degree to which the specified blob is planar, where 1.0 is a perfect plane and 0.0 is a perfect sphere. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_X	Uses the X-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_Y	Uses the Y-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_Z	Uses the Z-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_X	Uses the X-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_Y	Uses the Y-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_PRINCIPAL_AXIS_2_Z	Uses the Z-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_X	Uses the X-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Y	Uses the Y-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Z	Uses the Z-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX_ANGLE	Uses the angle of the semi-oriented bounding box's rotation around the Z-axis, measured from the X-axis to the box's longest XY-side, respecting the right-hand rule. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X +	Uses the size in X of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +	Uses the size in Y of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z +	Uses the size in Z of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_X	Uses the standard deviation of points in the blob along the X-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Y	Uses the standard deviation of points in the blob along the Y-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Z	Uses the standard deviation of points in the blob along the Z-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_BOX_CENTER_X; M_BOX_CENTER_Y; M_BOX_CENTER_Z; M_SIZE_X; M_SIZE_Y; M_SIZE_Z.

You can add one of the following values to the above-mentioned values to specify the feature's bounding box.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of bounding box																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_PCA_BOX	Specifies to use results associated with the PCA bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX	Specifies to use results associated with the semi-oriented bounding box.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER; M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to use a contact point of the Feret.

The calculation of Feret contact points (M_FERET_CONTACT_POINTS) must have been enabled to use these combination constants.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using Feret contact points																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_CONTACT_POINTS_X1	Uses the X-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_X2	Uses the X-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y1	Uses the Y-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y2	Uses the Y-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z1	Uses the Z-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z2	Uses the Z-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER; M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to use the direction of the Feret.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using the Feret direction																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_FERET_DIRECTION_X	Uses the X-component of the Feret's direction vector. For example, the direction of M_FERET_X is (1, 0, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Y	Uses the Y-component of the Feret's direction vector. For example, the direction of M_FERET_Y is (0, 1, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Z	Uses the Z-component of the Feret's direction vector. For example, the direction of M_FERET_Z is (0, 0, 1).	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_NEAREST_BLOB.

You can add one of the following values to the above-mentioned value to specify to use a coordinate of the closest point between the current blob and its nearest neighbor.

Note that, to use these combination values, M_NEAREST_BLOB cannot be equal to 0.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using the closest points between neighboring blobs																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NEAREST_POINT_X1	Uses the X-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_X2	Uses the X-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Y1	Uses the Y-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Y2	Uses the Y-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z1	Uses the Z-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z2	Uses the Z-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b

Condition

Specifies the condition that the specified SelectionCriterion must satisfy.

To base the selection on whether or not the specified feature result is available, set the Condition parameter to one of the values below. In either case, you must set CondLow and CondHigh to M_NULL.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For selection using a feature result's availability																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_AVAILABLE	Specifies to select for feature results that are available in the source result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_NOT_AVAILABLE	Specifies to select for feature results that are not available in the source result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

For conditions that use two limits, set the Condition parameter to one of the values below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For selection using two limits																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_IN_RANGE	Specifies to base the selection on the specified feature values that are in the range CondLow to CondHigh, inclusive .	c		e	f	g	a		i		j	k	l	m	n	b
M_OUT_RANGE	Specifies to base the selection on the specified feature values that are less than CondLow, or greater than CondHigh .	c		e	f	g	a		i		j	k	l	m	n	b

For conditions that use only one limit, set the Condition parameter to one of the values below. In all cases, you must set CondHigh to M_NULL.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For selection using only one limit																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_EQUAL	Specifies to base the selection on the specified feature values that are equal to CondLow .	c		e	f	g	a		i		j	k	l	m	n	b
M_GREATER	Specifies to base the selection on the specified feature values that are greater than CondLow .	c		e	f	g	a		i		j	k	l	m	n	b
M_GREATER_OR_EQUAL	Specifies to base the selection on the specified feature values that are greater than or equal to CondLow .	c		e	f	g	a		i		j	k	l	m	n	b
M_LESS	Specifies to base the selection on the specified feature values that are less than CondLow .	c		e	f	g	a		i		j	k	l	m	n	b
M_LESS_OR_EQUAL	Specifies to base the selection on the specified feature values that are less than or equal to CondLow .	c		e	f	g	a		i		j	k	l	m	n	b
M_NOT_EQUAL	Specifies to base the selection on the specified feature values that are not equal to CondLow .	c		e	f	g	a		i		j	k	l	m	n	b

CondLow

Specifies the lower limit of the selected condition.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the lower limit																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that no lower limit is required. This setting should be selected when Condition is set to M_AVAILABLE or M_NOT_AVAILABLE.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the lower limit.	c		e	f	g	a		i		j	k	l	m	n	b

CondHigh

Specifies the upper limit of the selected condition.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the upper limit																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that no upper limit is required. This setting should be selected for conditions that use only one limit, or when Condition is set to M_AVAILABLE or M_NOT_AVAILABLE.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the upper limit of the condition. This setting is required only when Condition is set to M_IN_RANGE or M_OUT_RANGE.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobSort

Synopsis

Sort the indices assigned to blobs based on their blob features, and set the destination result buffer to recognize the blobs using these indices; corresponding blob identifying information and feature results are copied to the specified destination result buffer.

Syntax

```
void M3dblobSort(
    AIL_ID SrcResult3dblobId, //in
    AIL_ID DstResult3dblobId, //out
    AIL_INT64 Feature1,       //in
    AIL_INT64 Feature2,       //in
    AIL_INT64 Feature3,       //in
    AIL_INT NbBlobsToKeep,    //in
    AIL_INT64 ControlFlag     //in
)
```

Description

This function sorts the indices assigned to blobs based on their blob features, and sets the destination result buffer to recognize the blobs using these indices; corresponding blob identifying information and feature results are copied to the specified destination result buffer. Sorting changes the blobs' indices; the labels remain unchanged. M3dblobSegment() must have been called at least once before calling this function.

By default, blob indices are sorted in ascending order. For example, the first index in the list is assigned to the blob whose calculated feature value is the smallest. Note that you can change the order (for example, to sort in descending order).

Note that, when passing the sorted result buffer to functions where you might specify an index (for example, M3dblobExtract()), the blob indices will have changed because of the sort. For other functions (such as M3dblobCombine()), the sorted result buffer is still compatible with other result buffers that come from the same blob segmentation (M3dblobSegment()).

This function supports in-place processing; you can specify the identifier of the same result buffer for the source and the destination.

Parameters

SrcResult3dblobId

Specifies the identifier of the source 3D blob analysis result buffer from which to sort the blob indices.

DstResult3dblobId

Specifies the identifier of the destination 3D blob analysis result buffer into which to copy the sorted indices, feature results, and other blob information. In-place processing is supported; DstResult3dblobId can refer to the same result buffer as SrcResult3dblobId.

Feature1

Specifies the first feature to use for sorting. This can be any of the following features; these features correspond to any non-array feature available using M3dblobGetResult() that is also retrievable using a specific blob label.

If a feature result is not available, ControlFlag determines the sorting for that blob's index.

For specifying the first feature																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_CUSTOM_FEATURE(AIL_INT N)	Uses the computed values of the specified custom feature that was manually added to the source result buffer using M3dblobControl() with M_CUSTOM_FEATURE(). This feature was not calculated using M3dblobCalculate(). You can add a combination value from the following table: • For specifying the sorting order.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	N Specifies the index of the custom feature.															
	0 <= *Value* < 1024 — Specifies the valid index of the custom feature.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_CENTRAL_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ)	Uses the central moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where *N* is the number of points in the blob, μ is the centroid, and (x_i, y_i, z_i) is the ith point: $\sum_i (x_i - \mu_x)^{PowerX} (y_i - \mu_y)^{PowerY} (z_i - \mu_z)^{PowerZ}$ You can add a combination value from the following table: • For specifying the sorting order.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PowerX Specifies the power in X.															
	Value >= 0 — Specifies the power.															
	PowerY Specifies the power in Y.															
	Value >= 0 — Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b

		PowerZ Specifies the power in Z.														
		Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ)		Uses the (ordinary, non-central) moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where *N* is the number of points in the blob and (x_i, y_i, z_i) is the ith point: $\sum_i x_i^{PowerX} y_i^{PowerY} z_i^{PowerZ}$ You can add a combination value from the following table: • For specifying the sorting order.	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters														
		PowerX Specifies the power in X.														
		Value >= 0 Specifies the power.														
		PowerY Specifies the power in Y.														
		Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
		PowerZ Specifies the power in Z.														
		Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_X +		Uses the X-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_Y +		Uses the Y-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_AVERAGE_NORMAL_Z +	Uses the Z-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_TOUCHING_IMAGE_BORDERS +	Uses information about which side(s) of the container or label image that the blob touches. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_X +	Uses the X-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_Y +	Uses the Y-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_Z +	Uses the Z-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_X +	Uses the X-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Y +	Uses the Y-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Z +	Uses the Z-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_1 +	Uses the eigenvalue corresponding to the first principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_2 +	Uses the eigenvalue corresponding to the second principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_3 +	Uses the eigenvalue corresponding to the third principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_1 +	Uses the length of the Feret along the first principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_2 +	Uses the length of the Feret along the second principal axis.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_AT_PRINCIPAL_AXIS_3 +	Uses the length of the Feret along the third principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL +	Uses the length of the Feret in the custom direction that was specified using M3dblobControl() with M_FERET_GENERAL + M_FERET_DIRECTION_....	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER +	Uses the length of the longest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_DIAMETER +	Uses the length of the shortest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_HEIGHT +	Uses the height (smallest XY side) of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_WIDTH +	Uses the width (longest XY side) of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_X +	Uses the length of the X-aligned Feret (same as M_SIZE_X).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Y +	Uses the length of the Y-aligned Feret (same as M_SIZE_Y).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Z +	Uses the length of the Z-aligned Feret (same as M_SIZE_Z).	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_X +	Uses the X-coordinate of the first pixel in the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_Y +	Uses the Y-coordinate of the first pixel in the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_INDEX_VALUE +	Uses the index of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_VALUE +	Uses the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINEARITY +	Uses the degree to which the specified blob is linear, where 1.0 is a perfect line and 0.0 is a perfect sphere. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_X +	Uses the X-coordinate of the maximum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_MAX_Y +	Uses the Y-coordinate of the maximum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Z +	Uses the Z-coordinate of the maximum corner of the blob's axis-aligned or semi-oriented bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_X +	Uses the X-coordinate of the minimum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Y +	Uses the Y-coordinate of the minimum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Z +	Uses the Z-coordinate of the minimum corner of the blob's axis-aligned or semi-oriented bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB +	Uses the label of the nearest blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB_DISTANCE +	Uses the minimum distance to the nearest blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS +	Uses the number of points in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_X +	Uses the maximum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_Y +	Uses the maximum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_X +	Uses the minimum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_Y +	Uses the minimum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b

M_PLANARITY +	Uses the degree to which the specified blob is planar, where 1.0 is a perfect plane and 0.0 is a perfect sphere. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_X +	Uses the X-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_Y +	Uses the Y-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_Z +	Uses the Z-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_X +	Uses the X-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_Y +	Uses the Y-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_Z +	Uses the Z-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_X +	Uses the X-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Y +	Uses the Y-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Z +	Uses the Z-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX_ANGLE +	Uses the angle of the semi-oriented bounding box's rotation around the Z-axis, measured from the X-axis to the box's longest XY-side, respecting the right-hand rule. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_X +	Uses the size in X of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +	Uses the size in Y of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z +	Uses the size in Z of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_X +	Uses the standard deviation of points in the blob along the X-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Y +	Uses the standard deviation of points in the blob along the Y-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Z +	Uses the standard deviation of points in the blob along the Z-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_BOX_CENTER_X; M_BOX_CENTER_Y; M_BOX_CENTER_Z; M_SIZE_X; M_SIZE_Y; M_SIZE_Z.

You can add one of the following values to the above-mentioned values to specify the feature's bounding box.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of bounding box																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_PCA_BOX	Specifies to use results associated with the PCA bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX	Specifies to use results associated with the semi-oriented bounding box.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER; M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to use a contact point of the Feret.

The calculation of Feret contact points (M_FERET_CONTACT_POINTS) must have been enabled to use these combination constants.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using Feret contact points

Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_CONTACT_POINTS_X1	Uses the X-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_X2	Uses the X-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y1	Uses the Y-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y2	Uses the Y-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z1	Uses the Z-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z2	Uses the Z-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER; M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to use the direction of the Feret.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using the Feret direction																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_DIRECTION_X	Uses the X-component of the Feret's direction vector. For example, the direction of M_FERET_X is (1, 0, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Y	Uses the Y-component of the Feret's direction vector. For example, the direction of M_FERET_Y is (0, 1, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Z	Uses the Z-component of the Feret's direction vector. For example, the direction of M_FERET_Z is (0, 0, 1).	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_NEAREST_BLOB.

You can add one of the following values to the above-mentioned value to specify to use a coordinate of the closest point between the current blob and its nearest neighbor.

Note that, to use these combination values, M_NEAREST_BLOB cannot be equal to 0.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using the closest points between neighboring blobs																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_NEAREST_POINT_X1	Uses the X-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_X2	Uses the X-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Y1	Uses the Y-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Y2	Uses the Y-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z1	Uses the Z-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z2	Uses the Z-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For specifying the first feature.

You can add one of the following values to the above-mentioned values to specify the sorting order of the blob indices.

The sort is based on the specified feature's calculated value. Use the ControlFlag parameter to modify index assignment to blobs for which the specified feature was not calculated.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the sorting order																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DOWN	Specifies to sort blob indices in descending order, from the largest calculated feature value to the smallest.	c		e	f	g	a		i		j	k	l	m	n	b
M_UP	Specifies to sort blob indices in ascending order, from the smallest calculated feature value to the largest. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

Feature2

Specifies the second feature to use for sorting, when the comparison using Feature1 is indeterminate. This feature can be any of the following.

If not required, set this parameter to **M_DEFAULT**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the second feature																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_CUSTOM_FEATURE(AIL_INT N)	Uses the computed values of the specified custom feature that was manually added to the source result buffer using M3dblobControl() with M_CUSTOM_FEATURE(). This feature was not calculated using M3dblobCalculate(). You can add a combination value from the following table: • For specifying the sorting order for the second feature.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	N Specifies the index of the custom feature.														
	0 <= *Value* < 1024 — Specifies the valid index of the custom feature.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_CENTRAL_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ)	Uses the central moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where *N* is the number of points in the blob, μ is the centroid, and (x_i, y_i, z_i) is the ith point: $\sum_i (x_i - \mu_x)^{PowerX} (y_i - \mu_y)^{PowerY} (z_i - \mu_z)^{PowerZ}$ You can add a combination value from the following table: • For specifying the sorting order for the second feature.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	PowerX Specifies the power in X.														
	Value >= 0 — Specifies the power.														
	PowerY Specifies the power in Y.														
	Value >= 0 — Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b

		PowerZ Specifies the power in Z.														
		Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ)		Uses the (ordinary, non-central) moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where *N* is the number of points in the blob and (x_i, y_i, z_i) is the ith point: $\sum_i x_i^{PowerX} y_i^{PowerY} z_i^{PowerZ}$ You can add a combination value from the following table: • For specifying the sorting order for the second feature.	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters														
		PowerX Specifies the power in X.														
		Value >= 0 Specifies the power.														
		PowerY Specifies the power in Y.														
		Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
		PowerZ Specifies the power in Z.														
		Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_X +		Uses the X-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_Y +		Uses the Y-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_AVERAGE_NORMAL_Z +	Uses the Z-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_TOUCHING_IMAGE_BORDERS +	Uses information about which side(s) of the container or label image that the blob touches. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_X +	Uses the X-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_Y +	Uses the Y-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_Z +	Uses the Z-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_X +	Uses the X-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Y +	Uses the Y-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Z +	Uses the Z-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_1 +	Uses the eigenvalue corresponding to the first principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_2 +	Uses the eigenvalue corresponding to the second principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_3 +	Uses the eigenvalue corresponding to the third principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_1 +	Uses the length of the Feret along the first principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_2 +	Uses the length of the Feret along the second principal axis.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_AT_PRINCIPAL_AXIS_3 +	Uses the length of the Feret along the third principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL +	Uses the length of the Feret in the custom direction that was specified using M3dblobControl() with M_FERET_GENERAL + M_FERET_DIRECTION_....	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER +	Uses the length of the longest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_DIAMETER +	Uses the length of the shortest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_HEIGHT +	Uses the height (smallest XY side) of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_WIDTH +	Uses the width (longest XY side) of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_X +	Uses the length of the X-aligned Feret (same as M_SIZE_X).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Y +	Uses the length of the Y-aligned Feret (same as M_SIZE_Y).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Z +	Uses the length of the Z-aligned Feret (same as M_SIZE_Z).	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_X +	Uses the X-coordinate of the first pixel in the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_Y +	Uses the Y-coordinate of the first pixel in the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_INDEX_VALUE +	Uses the index of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_VALUE +	Uses the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINEARITY +	Uses the degree to which the specified blob is linear, where 1.0 is a perfect line and 0.0 is a perfect sphere. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_X +	Uses the X-coordinate of the maximum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_MAX_Y +	Uses the Y-coordinate of the maximum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Z +	Uses the Z-coordinate of the maximum corner of the blob's axis-aligned or semi-oriented bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_X +	Uses the X-coordinate of the minimum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Y +	Uses the Y-coordinate of the minimum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Z +	Uses the Z-coordinate of the minimum corner of the blob's axis-aligned or semi-oriented bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB +	Uses the label of the nearest blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB_DISTANCE +	Uses the minimum distance to the nearest blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS +	Uses the number of points in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_X +	Uses the maximum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_Y +	Uses the maximum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_X +	Uses the minimum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_Y +	Uses the minimum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b

M_PLANARITY +	Uses the degree to which the specified blob is planar, where 1.0 is a perfect plane and 0.0 is a perfect sphere. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_X +	Uses the X-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_Y +	Uses the Y-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_Z +	Uses the Z-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_X +	Uses the X-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_Y +	Uses the Y-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_Z +	Uses the Z-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_X +	Uses the X-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Y +	Uses the Y-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Z +	Uses the Z-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX_ANGLE +	Uses the angle of the semi-oriented bounding box's rotation around the Z-axis, measured from the X-axis to the box's longest XY-side, respecting the right-hand rule. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_X +	Uses the size in X of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +	Uses the size in Y of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z +	Uses the size in Z of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_X +	Uses the standard deviation of points in the blob along the X-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Y +	Uses the standard deviation of points in the blob along the Y-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Z +	Uses the standard deviation of points in the blob along the Z-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_BOX_CENTER_X; M_BOX_CENTER_Y; M_BOX_CENTER_Z; M_SIZE_X; M_SIZE_Y; M_SIZE_Z.

You can add one of the following values to the above-mentioned values to specify the feature's bounding box.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of bounding box for the second feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_PCA_BOX	Specifies to use results associated with the PCA bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX	Specifies to use results associated with the semi-oriented bounding box.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER; M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to use a contact point of the Feret.

The calculation of Feret contact points (M_FERET_CONTACT_POINTS) must have been enabled to use these combination constants.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using Feret contact points for the second feature

Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_CONTACT_POINTS_X1	Uses the X-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_X2	Uses the X-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y1	Uses the Y-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y2	Uses the Y-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z1	Uses the Z-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z2	Uses the Z-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER; M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to use the direction of the Feret.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using the Feret direction for the second feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_DIRECTION_X	Uses the X-component of the Feret's direction vector. For example, the direction of M_FERET_X is (1, 0, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Y	Uses the Y-component of the Feret's direction vector. For example, the direction of M_FERET_Y is (0, 1, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Z	Uses the Z-component of the Feret's direction vector. For example, the direction of M_FERET_Z is (0, 0, 1).	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_NEAREST_BLOB.

You can add one of the following values to the above-mentioned value to specify to use a coordinate of the closest point between the current blob and its nearest neighbor.

Note that, to use these combination values, M_NEAREST_BLOB cannot be equal to 0.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using the closest points between neighboring blobs for the second feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_NEAREST_POINT_X1	Uses the X-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_X2	Uses the X-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Y1	Uses the Y-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Y2	Uses the Y-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z1	Uses the Z-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z2	Uses the Z-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For specifying the second feature.

You can add one of the following values to the above-mentioned values to specify the sorting order of the blob indices.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the sorting order for the second feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DOWN	Specifies to sort blob indices in descending order, from the largest calculated feature value to the smallest.	c		e	f	g	a		i		j	k	l	m	n	b
M_UP	Specifies to sort blob indices in ascending order, from the smallest calculated feature value to the largest.	c		e	f	g	a		i		j	k	l	m	n	b

Feature3

Specifies the third feature to use for sorting, when the comparisons using Feature1 and Feature2 are indeterminate. This feature can be any of the following.

If not required, set this parameter to **M_DEFAULT**.

> This parameter must be set to **M_DEFAULT** if Feature2 is set to **M_DEFAULT**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the third feature																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_CUSTOM_FEATURE(AIL_INT N)	Uses the computed values of the specified custom feature that was manually added to the source result buffer using M3dblobControl() with M_CUSTOM_FEATURE(). This feature was not calculated using M3dblobCalculate(). You can add a combination value from the following table: • For specifying the sorting order for the third feature.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	N Specifies the index of the custom feature.														
	0 <= *Value* < 1024 — Specifies the valid index of the custom feature.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_CENTRAL_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ)	Uses the central moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where *N* is the number of points in the blob, μ is the centroid, and (x_i, y_i, z_i) is the ith point: $\sum_i (x_i - \mu_x)^{PowerX} (y_i - \mu_y)^{PowerY} (z_i - \mu_z)^{PowerZ}$ You can add a combination value from the following table: • For specifying the sorting order for the third feature.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	PowerX Specifies the power in X.														
	Value >= 0 — Specifies the power.														
	PowerY Specifies the power in Y.														
	Value >= 0 — Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b

	PowerZ Specifies the power in Z.															
	Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_XYZ(AIL_INT PowerX, AIL_INT PowerY, AIL_INT PowerZ)	Uses the (ordinary, non-central) moment corresponding to the specified powers of X, Y, and Z. This is equivalent to the following formula, where *N* is the number of points in the blob and (x_i, y_i, z_i) is the ith point: $\sum_i x_i^{PowerX} y_i^{PowerY} z_i^{PowerZ}$ You can add a combination value from the following table: • For specifying the sorting order for the third feature.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PowerX Specifies the power in X.															
	Value >= 0 Specifies the power.															
	PowerY Specifies the power in Y.															
	Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
	PowerZ Specifies the power in Z.															
	Value >= 0 Specifies the power.	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_X +	Uses the X-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_AVERAGE_NORMAL_Y +	Uses the Y-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_AVERAGE_NORMAL_Z +	Uses the Z-component of the blob's average normal vector. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BLOB_TOUCHING_IMAGE_BORDERS +	Uses information about which side(s) of the container or label image that the blob touches. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_X +	Uses the X-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_Y +	Uses the Y-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_CENTER_Z +	Uses the Z-coordinate of the center of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination value M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_X +	Uses the X-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Y +	Uses the Y-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTROID_Z +	Uses the Z-coordinate of the centroid of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_1 +	Uses the eigenvalue corresponding to the first principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_2 +	Uses the eigenvalue corresponding to the second principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EIGENVALUE_3 +	Uses the eigenvalue corresponding to the third principal axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_1 +	Uses the length of the Feret along the first principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_AT_PRINCIPAL_AXIS_2 +	Uses the length of the Feret along the second principal axis.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_AT_PRINCIPAL_AXIS_3 +	Uses the length of the Feret along the third principal axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL +	Uses the length of the Feret in the custom direction that was specified using M3dblobControl() with M_FERET_GENERAL + M_FERET_DIRECTION_....	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER +	Uses the length of the longest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_DIAMETER +	Uses the length of the shortest Feret in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_HEIGHT +	Uses the height (smallest XY side) of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_SEMI_ORIENTED_WIDTH +	Uses the width (longest XY side) of the semi-oriented box.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_X +	Uses the length of the X-aligned Feret (same as M_SIZE_X).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Y +	Uses the length of the Y-aligned Feret (same as M_SIZE_Y).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Z +	Uses the length of the Z-aligned Feret (same as M_SIZE_Z).	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_X +	Uses the X-coordinate of the first pixel in the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_PIXEL_Y +	Uses the Y-coordinate of the first pixel in the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_INDEX_VALUE +	Uses the index of the blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_VALUE +	Uses the label of the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINEARITY +	Uses the degree to which the specified blob is linear, where 1.0 is a perfect line and 0.0 is a perfect sphere. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_X +	Uses the X-coordinate of the maximum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Y +	Uses the Y-coordinate of the maximum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_MAX_Z +	Uses the Z-coordinate of the maximum corner of the blob's axis-aligned or semi-oriented bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_X +	Uses the X-coordinate of the minimum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Y +	Uses the Y-coordinate of the minimum corner of the blob's axis-aligned bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Z +	Uses the Z-coordinate of the minimum corner of the blob's axis-aligned or semi-oriented bounding box. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB +	Uses the label of the nearest blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_BLOB_DISTANCE +	Uses the minimum distance to the nearest blob. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS +	Uses the number of points in the blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_X +	Uses the maximum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MAX_Y +	Uses the maximum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_X +	Uses the minimum X-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL_MIN_Y +	Uses the minimum Y-coordinate of the blob's stored location in the container's 2D grid organization (or the equivalent coordinate in the label image), expressed in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANARITY +	Uses the degree to which the specified blob is planar, where 1.0 is a perfect plane and 0.0 is a perfect sphere. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_PRINCIPAL_AXIS_1_X +	Uses the X-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_Y +	Uses the Y-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_1_Z +	Uses the Z-component of the normalized vector along the first principal axis, which is the axis with the largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_X +	Uses the X-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_Y +	Uses the Y-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_2_Z +	Uses the Z-component of the normalized vector along the second principal axis, which is the axis with the second largest eigenvalue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_X +	Uses the X-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Y +	Uses the Y-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRINCIPAL_AXIS_3_Z +	Uses the Z-component of the normalized vector along the third principal axis, which is the axis perpendicular to the first and second principal axes. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX_ANGLE +	Uses the angle of the semi-oriented bounding box's rotation around the Z-axis, measured from the X-axis to the box's longest XY-side, respecting the right-hand rule. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X +	Uses the size in X of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_Y +	Uses the size in Y of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z +	Uses the size in Z of the blob's bounding box. The default bounding box is axis-aligned. To specify the PCA or semi-oriented box, use the respective combination values M_PCA_BOX or M_SEMI_ORIENTED_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_X +	Uses the standard deviation of points in the blob along the X-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Y +	Uses the standard deviation of points in the blob along the Y-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_STANDARD_DEVIATION_Z +	Uses the standard deviation of points in the blob along the Z-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_BOX_CENTER_X; M_BOX_CENTER_Y; M_BOX_CENTER_Z; M_SIZE_X; M_SIZE_Y; M_SIZE_Z.

You can add one of the following values to the above-mentioned values to specify the feature's bounding box.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of bounding box for the third feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_PCA_BOX	Specifies to use results associated with the PCA bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEMI_ORIENTED_BOX	Specifies to use results associated with the semi-oriented bounding box.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER; M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to use a contact point of the Feret.

The calculation of Feret contact points (M_FERET_CONTACT_POINTS) must have been enabled to use these combination constants.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using Feret contact points for the third feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_CONTACT_POINTS_X1	Uses the X-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_CONTACT_POINTS_X2	Uses the X-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y1	Uses the Y-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Y2	Uses the Y-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z1	Uses the Z-coordinate of the start point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_CONTACT_POINTS_Z2	Uses the Z-coordinate of the end point of the blob's Feret diameter.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_AT_PRINCIPAL_AXIS_1; M_FERET_AT_PRINCIPAL_AXIS_2; M_FERET_AT_PRINCIPAL_AXIS_3; M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER; M_FERET_SEMI_ORIENTED_HEIGHT; M_FERET_SEMI_ORIENTED_WIDTH; M_FERET_X; M_FERET_Y; M_FERET_Z.

You can add one of the following values to the above-mentioned values to specify to use the direction of the Feret.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using the Feret direction for the third feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FERET_DIRECTION_X	Uses the X-component of the Feret's direction vector. For example, the direction of M_FERET_X is (1, 0, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Y	Uses the Y-component of the Feret's direction vector. For example, the direction of M_FERET_Y is (0, 1, 0).	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_DIRECTION_Z	Uses the Z-component of the Feret's direction vector. For example, the direction of M_FERET_Z is (0, 0, 1).	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_NEAREST_BLOB.

You can add one of the following values to the above-mentioned value to specify to use a coordinate of the closest point between the current blob and its nearest neighbor.

Note that, to use these combination values, M_NEAREST_BLOB cannot be equal to 0.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For using the closest points between neighboring blobs for the third feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NEAREST_POINT_X1	Uses the X-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_X2	Uses the X-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b

M_NEAREST_POINT_Y1	Uses the Y-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Y2	Uses the Y-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z1	Uses the Z-coordinate of the point inside the current blob that is closest to the nearest neighboring blob.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_POINT_Z2	Uses the Z-coordinate of the point inside the nearest neighboring blob that is closest to the current blob.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For specifying the third feature.

You can add one of the following values to the above-mentioned values to specify the sorting order of the blob indices.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the sorting order for the third feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DOWN	Specifies to sort blob indices in descending order, from the largest calculated feature value to the smallest.	c		e	f	g	a		i		j	k	l	m	n	b
M_UP	Specifies to sort blob indices in ascending order, from the smallest calculated feature value to the largest.	c		e	f	g	a		i		j	k	l	m	n	b

NbBlobsToKeep

Specifies to keep the first N blobs, if required. To keep all blobs, set this to M_INFINITE or M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which blobs to keep																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies to keep all blobs.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of blobs to keep.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Specifies how to deal with blob indices for which the blob's specified feature was not calculated. This is only applied when necessary. For example, if both blobs have a Feature1 result and the comparison is successful, then whether or not both blobs have a Feature2 result is irrelevant, and the ControlFlag setting is ignored.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For dealing with blob indices for which the blob's specified feature was not calculated

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_FEATURE_REQUIRED.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACK	Specifies to put blob indices, for blobs that do not have the feature result, at the end of the list.	c		e	f	g	a		i		j	k	l	m	n	b
M_FEATURE_REQUIRED	Specifies that a calculated feature result must exist for the specified feature; otherwise, an error is reported.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRONT	Specifies to put blob indices, for blobs that do not have the feature result, at the front of the list.	c		e	f	g	a		i		j	k	l	m	n	b
M_GREATER	Specifies that blobs that do not have the feature result compare larger than blobs that do. This means that, when comparing two blobs (one with and one without the feature result), the missing feature result's value is considered larger for that blob, which affects the blob's index placement in the list. For example, if the sort is ascending (M_UP), the blob without the feature result is placed in the list after the blob that has the feature result.	c		e	f	g	a		i		j	k	l	m	n	b
M_LESS	Specifies that blobs that do not have the feature result compare smaller than blobs that do. This means that, when comparing two blobs (one with and one without the feature result), the missing feature result's value is considered smaller for that blob, which affects the blob's placement in the list. For example, if the sort is ascending (M_UP), the blob without the feature result is placed in the list before the blob that has the feature result.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.

M3dblobStream

Synopsis

Load, restore, or save a 3D blob analysis context from/to a file or memory stream.

Syntax

```
void M3dblobStream(
    AIL_TEXT_PTR MemPtrOrFileName, //in-out
    AIL_ID SysId,                  //in
    AIL_INT64 Operation,           //in
    AIL_INT64 StreamType,          //in
    AIL_DOUBLE Version,            //in
    AIL_INT64 ControlFlag,         //in
    AIL_ID *Context3dblobIdPtr,    //in-out
    AIL_INT *SizeByteVarPtr        //out
)
```

Description

This function can load, restore, or save a 3D blob analysis context from/to a file or memory stream.

To inquire the number of bytes necessary to save a 3D blob analysis context to a memory stream, you should first call this function (**M3dblobStream**()) with M_INQUIRE_SIZE_BYTE.

The content saved to memory stream is equivalent to the content saved to file. In addition, any file saved using this function is equivalent to a file saved with M3dblobSave().

You can use this and other Aurora Imaging Library stream functions, for example, to save all required Aurora Imaging Library objects, as well as any other custom data, for your application to a memory stream. Once in a memory stream, you can write the stream to a single file or transfer it over a network. You are responsible for concatenating the streams and for saving the stream to file.

This function restores all of the object's settings that were in effect when the object was saved.

Using **M3dblobStream**(), you can choose to save a backwards-compatible version of the 3D blob analysis context, which will work using a version of Aurora Imaging Library that is up to one major release older than the current version (depending on which version is specified). However, all settings and features unique to the higher version will be ignored when restored using the lower version. Besides saving backwards-compatible versions, you can also load or restore 3D blob analysis contexts saved using any Aurora Imaging Library version with which this function was available. Settings that do not exist in the lower version will be filled with default values when the 3D blob analysis context is loaded or restored.

Parameters

MemPtrOrFileName
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_TEXT_PTR [optionally, in C++: a reference to an AIL_STRING]
- Optionally, in C++: a reference to a std::vector of type AIL_UINT8

Specifies the file or memory stream.

See the Parameter associations section for possible values that can be returned/specified.

SysId

Specifies the system on which to restore the 3D blob analysis context.

See the Parameter associations section for possible values that can be specified.

Operation

Specifies the operation to perform.

See the Parameter associations section for possible values that can be specified.

StreamType

Specifies the type of stream in which to store/from which to restore the 3D blob analysis context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of stream																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILE	Specifies a file stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_MEMORY	Specifies a memory stream. You are responsible for allocating a block of memory for the stream.	c		e	f	g	a		i		j	k	l	m	n	b

Version

Specifies the Aurora Imaging Library version of the 3D blob analysis context.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

Context3dblobIdPtr

Specifies the address of the variable in which to write or from which to read the identifier of the 3D blob analysis context. If the M_RESTORE operation is not successful, **M_NULL** is returned.

See the Parameter associations section for possible values that can be returned/specified.

SizeByteVarPtr

Specifies the address of the variable in which to write the size of the 3D blob analysis context, in bytes.

If the size is not required, you can set this parameter to **M_NULL**.

> Note that the size of the 3D blob analysis context will vary depending on the Aurora Imaging Library version specified.

Parameter associations

The table below lists possible values for the **MemPtrOrFileName**, **SysId**, **Operation**, **Version**, and **Context3dblobIdPtr** parameters.

- For performing the stream operation.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For performing the stream operation.																
Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

MemPtrOrFileName - Possible values passed or returned																
SysId																
Version																
Context3dblobIdPtr - Possible values passed or returned																
M_INQUIRE_SIZE_BYTE	Inquires the number of bytes required to save a 3D blob analysis context to memory stream. This operation is not supported when the StreamType parameter is set to M_FILE.															
MemPtrOrFileName	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	Specifies the Aurora Imaging Library version with which the 3D blob analysis context must be compatible when saved. The size of a 3D blob analysis context might differ from one Aurora Imaging Library version to another.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP5	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 5.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
Context3dblobIdPtr	Specifies the address of the variable containing the identifier of the 3D blob analysis context.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOAD	Loads the content of a specified file or memory stream into a previously allocated 3D blob analysis context.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream from which to load the 3D blob analysis context.	c		e	f	g	a		i		j	k	l	m	n	b

		M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
		AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. 3D blob analysis contexts typically have an M3DBLOB file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
		MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
	SysId		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
	Version		This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
	Context3dblobIdPtr		Specifies the address of the variable containing the identifier of the 3D blob analysis context. All previous settings are overwritten with those of the loaded 3D blob analysis context, or are set to the default value if the settings didn't exist in the version of Aurora Imaging Library from which the 3D blob analysis context is being loaded.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESTORE			Restores a 3D blob analysis context from a file or memory stream and assigns it an Aurora Imaging Library identifier.	c		e	f	g	a		i		j	k	l	m	n	b
	MemPtrOrFileName		Specifies the file or memory stream from which to restore the 3D blob analysis context.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b

AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. 3D blob analysis contexts typically have an M3DBLOB file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	Specifies the system on which to allocate the 3D blob analysis context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
Version	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
Context3dblobIdPtr	Specifies the address of the variable in which to write the identifier of the 3D blob analysis context. If the operation is not successful, **M_NULL** is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE	Saves a 3D blob analysis context to a specified file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b

AIL_TEXT("*FileName*") [1]	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. For easier use with other Aurora Imaging software products, when saving a 3D blob analysis context to a file, use the M3DBLOB file extension. The function handles (internally) the opening and closing of the file. If the file already exists, it will be overwritten. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which to write, and the block of memory must be large enough to stream the entire context. To determine the required size, call this function with M_INQUIRE_SIZE_BYTE. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	Specifies the Aurora Imaging Library version with which the 3D blob analysis context must be compatible when saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP5	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 5.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
Context3dblobIdPtr	Specifies the address of the variable containing the identifier of the 3D blob analysis context to save.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dblob11.lib.
DLL	Requires ail11.dll; ail3dblob11.dll.